



A novel historical data-based method for predicting firefighters demand in urban fires

Chen-yue Zhang, Rui Zhao^{*}, Ning Wang, Xin Nie

School of Economics and Management, Dalian University of Technology, Dalian, 116024, China

ARTICLE INFO

Keywords:

Urban fires
Firefighters demand predict
Firefighters dispatch
Historical data-based
Artificial neural network

ABSTRACT

Urban fires are a prevalent type of fire that requires prompt and effective responses. Determining the appropriate number of firefighters at the alarm time is crucial, but it mostly relies on the experience of decision-makers. Due to the fuzziness and limitations of human knowledge, there is often a deviation between the number of firefighters dispatched relying on personal subjective experience and the realistic demand. This paper proposed a historical data-based method for predicting the demand number of firefighters in urban fire. Initially, the National Fire Incident Reporting System (NFIRS) data and Global Historical Climatology Network Daily (GHCN-D) weather data were combined to create a fused dataset. The dataset was subjected to anomaly detection and feature selection, then the processed data was used to predict the number of firefighters using artificial neural network (ANN). In this process, the Genetic Algorithm (GA) was applied to optimize ANN structure. Finally, comparative experiments were conducted to evaluate the performance of the proposed method, which demonstrated superior accuracy in comparison to common regression models and traditional ANN. This advancement brings the number of dispatched firefighters closer to the actual demand and contributes to ensuring an adequate allocation of firefighters for urban fires. Consequently, decision-makers can make more informed decisions regarding the number of firefighters to dispatch, leading to more effective responses to urban fires.

1. Introduction

With complex building structures, diverse functions and dense populations, cities are highly susceptible to fire accidents that can result in significant property damage and casualties [1]. Therefore, it is crucial to minimize such losses through various means. The initial stage of urban fire typically involves a small fire with relatively low temperatures, presenting an excellent opportunity for suppression. Consequently, the ability to dispatch an adequate number of firefighters during the initial stage can greatly improve the speed of firefighting [2].

Previous studies of urban fires have mainly focused on fire risk analysis and the allocation of fire forces, these were relatively abstract and macro [3,4]. However, there has been a lack of research on determining the specific number of firefighters demand, which is crucial to effectively managing actual fire incidents. Because decision-makers have limited information regarding the severity of the fire in emergency situations, they often struggle to determine the appropriate number of firefighters to dispatch. Dispatching too many firefighters can lead to a waste of human resources and the surplus firefighters may not

be available to respond to other emergency incidents. On the other hand, if the number of dispatched firefighters is insufficient, the best suppression chance may be missed.

To provide executive standards for firefighting and reduce the reliance on the experience of decision-makers, several corresponding response strategies have been proposed by federal and local fire departments. For instance, NFPA 1720 standard establishes a baseline for tasks in residential fire situations, including having at least one support person for each attack, at least one rescue team with a minimum of two members per team, and so on [5]. However, the strategies outlined in these documents are often vague and abstract. They cannot make specific recommendations on the number of firefighters demanded for each urban fire. Consequently, dispatching strategies that rely solely on experience are limited in their flexibility to adapt to complex situations. To the contrary, data-based prediction can extract useful information from historical data and provide precise number for firefighters dispatch. This is important for effectively addressing practical urban firefighters demand prediction problems.

Nowadays, the data-based method is widely used to provide decision

^{*} Corresponding author.

E-mail address: rzhao@mail.dlut.edu.cn (R. Zhao).

<https://doi.org/10.1016/j.firesaf.2024.104200>

Received 5 January 2024; Received in revised form 2 June 2024; Accepted 4 June 2024

Available online 6 June 2024

0379-7112/© 2024 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

support in various areas of emergency management [6,7]. However, there is limited research on firefighters demand prediction based on historical data. Additionally, none of the existing demand prediction methods take into account weather factors, which can significantly impact fire behavior. Weather variables such as temperature and wind speed can affect the spread and burning speed of fires [8–10]. Based on the above situations, this paper uses fire incidents data from the National Fire Incident Reporting System (NFIRS) and weather data from the Global Historical Climatology Network Daily (GHCN-D) to obtain basic information. To ensure the accuracy of the proposed method, only the features that can be obtained at the alarm time are selected as primary features. Subsequently, four popular anomaly detection methods (MCD, ABOD, Auto Encoder and Iforest) are respectively combined with four feature selection methods (XGBoost, GBDT, Random Forest and Ridge) to preprocess the dataset. This method aims to minimize the influence of special fire incidents and identify relevant features. Given that the artificial neural network (ANN) is excellent in handling nonlinear issues, this paper uses it to predict the processed data. Furthermore, genetic algorithm (GA) is utilized to optimize the ANN structure for reducing errors and improving the accuracy of demand prediction for firefighters.

The rest of this paper is organized as follows. Section 2. Provides the literature review, which includes research on response demand and dispatch of firefighters, data-based method for personnel demand and fire response, and urban firefighters demand prediction based on ANN. Section 3. is the method of this paper, includes the data preprocessing process and the construction of urban firefighters demand prediction model. This model utilizes Artificial Neural Network (ANN) optimized by Genetic Algorithm (GA) to estimate the number of firefighters demanded. Section 4. is dedicated to discussing the results, including the validation and evaluation of the method through analysis and comparison. Finally, Section 5. concludes this paper.

2. Literature review

2.1. Firefighters response demand and dispatch

Over the past few decades, extensive experience-based research has been conducted on the firefighters response demand and dispatch. Wybo proposed a decision support system for forest fires, utilizing an expert system model to identify some key indicators. This method assisted fire managers in emergency decision-making, which in turn helped with the dispatch of firefighters [11]. Njå et al. revealed the safety of Norwegian emergency responders before and after incidents by analyzing specific cases of tunnel fires in Norway, and gathering knowledge data from a workshop involving tunnel fire response experts [12]. Gazzard et al. studied wildfire response demand policies in the UK over the last two decades, analyzing wildfire management mechanisms at different regional, local and national levels [13]. Rezaeifam et al. conducted semi-structured situational interviews with 10 municipal fire department experts to determine the goals, decision-making and information demands of firefighters [14]. In addition, the discussion of fire station distribution is essentially an extension to study personnel dispatch. After evaluating existing fire stations, Yeboah et al. found that more stations should be established to increase coverage and reduce response times [15]. Furthermore, the dispatch system of firefighters as first responders can also reduce medical emergency response time to a certain extent [16].

Above studies explored the issue of firefighters response demand through the analysis of specific fire incidents, geographic information, the priori knowledge of experts, and the explanation of relevant strategies. These efforts fully illustrated the importance and significance of firefighters dispatch. Firefighters dispatch aims to save as many lives as possible with limited emergency response resources. Dispatching excessive firefighters would result in higher costs for emergency response including the wastage of human and material resources. Conversely, dispatching insufficient firefighters would lead to

inadequate emergency response and untimely dispatch, which will ultimately cause increased casualties and property damage, resulting in irreversible consequences.

The timely response of firefighters to fire outbreaks can significantly reduce the consequences of such incidents [17]. But dispatching suitable firefighters according to the demand is a difficult challenge in present. Urban fires are influenced by various factors. While personal experience and expert knowledge have limitations and biases, making it difficult to dispatch suitable number of firefighters in decision-making. Relying solely on experience tends to yield inaccurate and unstable results, which is not conducive to effective firefighters dispatch.

2.2. Data-based method for personnel demand and fire response

The application of data-based method to the field of personnel demand prediction has become a mainstream research trend as data science has advanced significantly. The method of predicting personnel demand based on historical data is more convincing and efficient than relying on subjective experience and expert knowledge. A growing number of personnel demand studies such as construction engineering, transportation and healthcare services have sought historical data-based method to predicting personnel demand. Using a stepwise regression model, Kim et al. implemented personnel hour prediction based on a dataset containing 6928 projects constructed between 2001 and 2011. This method aimed to dispatch an appropriate number of construction engineering personnel to effectively cover the extent of the district [18]. Li et al. combined qualitative literature research and quantitative data analysis to support the validation of the staffing requirements model [19]. Pang et al. collected historical weekly data on 90 beds for each section and utilized statistical models to predict hospital laboratory workload [20]. Using real data from the emergency department, Chen et al. verified the effectiveness of the random search heuristic algorithm based on simulated annealing. This method optimized the allocation of medical staff in the emergency department [21].

As evidenced by previous research, the integration of real data with statistical method and regression model has been widely used in personnel demand predicting. These methods have yielded significant results, allowing relevant practitioners to efficiently schedule personnel, thereby improving efficiency and reducing costs. However, it is important to note that the algorithmic logic of statistical method and regression model are relatively straightforward and relies on high-quality data. At the same time, these methods frequently struggle to effectively handle highly nonlinear data, leading to potential errors in predicting personnel demand.

Currently, the fire service is undergoing a modernization process that requires the rational utilization of a significant volume of historical data generated by urban fires. It is crucial to effectively mine the dispatch regularity of firefighters contained in historical data to support decision-making among the relevant departments. With the advancements in data science based on historical data-based method, there are now numerous traditional methods available for studying fire behavior and response. Further, the prediction of personnel demands is also part of the fire response study.

In the early stages about fire response, simulation method was widely used for modeling firefighting operations. As a notable achievement in the successful project for the Fire Department of New York (FDNY) [22,23], Carter et al. developed a simulation model to analyze the responses of firefighting units. This model took various factors into consideration, aiming to improve traditional dispatching strategies [24,25]. Following this, Ignall et al. further developed a simulation-based algorithm for initial fire response dispatch in New York City. They also verified that the model was effective in reducing dispatch times by comparing it to actual cases and policies [26]. Another commonly used method in this field is the Markov process. Swersey proposed a Markovian decision model to estimate the severity of the fire and the probability of false alarms at the alarm time. This method led to

significant improvements in response time to serious fires [27]. More recently, considering the problems of dispatching firefighters due to the inherent complexities in the environment, Mukhopadhyay et al. formulated the firefighters dispatch problem as a Semi-Markov decision process and built a decision theoretic framework for it. This framework effectively leveraged state-of-the-art methods for the prediction of continuous-time spatio-temporal incidents [28]. With the development of smart interconnection, a dispatch algorithm under an online mechanism to reduce dispatch time was subsequently proposed [29]. In a similar vein, Usanov et al. modeled the scheduling problem as a Markov process, which investigating the trade-off between coverage and response time in the dispatch of fire trucks [30].

Besides simulation methods and Markov processes, other data-based methods that have been used for the field of fire response. Base on the data of U.S. National Fire Incident Reporting System (NFIRS) from 2010 to 2015, Arnsbarger et al. modeled a generalized linear model for turnout time and a linear model for travel time respectively. These models helped to better understand the impact of various factors of structure fires on response times [31]. Dunn et al. developed a response framework for large wildfires by constructing an operations research model [32]. Take the advantage of GIS technology, Ishida et al. implemented a volunteer fire corps activity support system to help firefighters pinpoint target locations and improve rescue efficiency [33]. Additionally, the case-based reasoning (CBR) method is widely used in emergency resource predicting. For example, Basu et al. utilized the CBR technique to validate the feasibility of emergency resource requirements predicted by the principal component regression model [34]. As a traditional data-based method, CBR is often applied to fire management [35] and predict resource requirements [36], which are also relevant to fire response.

From the objective perspective, data-based method is more convincing and better equipped to handle fire response in complex contexts compared to experience-based method. However, traditional data-based methods such as probabilistic statistical methods, simulation models, and Markov processes rely on many rigorous mathematical assumptions and require high-quality historical data. On the other hand, spatial analysis method such as GIS uses a single dimension of data and can only extract limited information. CBR method is constrained by the unique characteristics of different cases, making it challenging to find a suitable case retrieval algorithm. In summary, all of these traditional data-based methods for analyzing fire response have limitations. Meanwhile, these methods are difficult to adapt to the complex and variational needs of urban fire scenarios.

In order to ensure the efficiency and precision of firefighters dispatch in the process of urban fires emergency response, it is usually necessary to predict the number of firefighters demanded to perform firefighting tasks based on historical data of urban fires, and rationally arrange the dispatch of firefighters accordingly. It is worth noting that complex and diverse factors influence urban fires, such as building structures and environmental conditions. Therefore, predicting firefighters demand based on historical data is exceedingly challenging. Additionally, the variability in the quality of historical data further constrains this area of research. Consequently, there is rarely method currently available that can predict the number of firefighters demanded based on historical data related to urban fires. Scholars often focus on discussing ways to reduce dispatch time instead. Therefore, it is significant potential for conducting further in-depth research on the appropriate number of firefighters demanded for urban fires based on historical data.

2.3. Urban firefighters demand prediction based on ANN

Machine learning (ML) method is very popular for predicting personnel demand, as they leverage historical data to provide quick and accurate predictions. With the development of computer technology, ML has introduced some new solutions for addressing the complex problem of personnel demand predicting, including time series models,

optimization models and so on [37]. Among these ML methods, artificial neural network (ANN) was widely used and considered more mature with better performance, accuracy and robustness. Elkhooly et al. have come up with a general model that took into account a multilayer perceptron artificial neural network with back propagation to predict labor requirements [38]. There is also the radial basis function neural network [39], the deep neural network and reinforcement learning [40], which have been used to predict personnel demand or workforce allocation.

On the other hand, ANN has been successfully applied to emergency management [41–45], including firefighters demand prediction. The use of ANN has demonstrated strong predictive power in accurately determining the number of firefighters demanded for different fire incidents. For illustration, Sadeghi proposed a feed-forward backpropagation neural network to model the number of firefighters dispatched to different fire incidents, which reported highly promising abilities [46]. These findings highlight the potential of artificial neural network for improving firefighters demand prediction processes. By leveraging advanced algorithms and training on historical data, artificial neural network can provide valuable insights and assist in making informed decisions regarding the allocation of firefighters.

To further enhance the predictive ability of artificial neural network, there are some mainstream optimization methods. It's common to use gradient descent, stochastic gradient descent and the improved algorithms based on these methods (e.g., Adam [47], AdaDelta [48], RMSprop [49]) to optimize the parameters of ANN. In addition to the above optimization algorithms on the basis of numerical analysis, evolutionary algorithms are also commonly used in the ANN optimization process, such as grey wolf optimization [50] and the beetle search algorithm [51]. The genetic algorithm (GA) is an evolutionary algorithm for heuristic stochastic search, known for its strong global searching ability [52]. Combining ANN with GA provides the potential for obtaining better demand prediction models, as demonstrated in various fields and studies [53].

In view of the above, ANN is widely used in the fields of firefighters demand prediction. Various algorithms effectively enhance the performance of ANN. However, the existing optimization of ANN model is more inclined to optimize the model parameters and ignores the optimization of the overall structure. It is worth noting that when optimizing ANN model, not only the model parameters should be optimized, but also the structure of the model is needed as well. This can improve the prediction performance and computation efficiency of ANN [54]. Therefore, to get more accurate results of firefighters demand prediction, this paper tried to optimize both the parameters and structure of ANN. Considering the balance between efficiency and accuracy, NAdam algorithm was used during the training process to optimize parameters. This algorithm is clearly outperformed Adam in reducing training and verification losses [55]. In addition, GA was utilized to optimize the ANN structure.

3. Methods

The proposed method to predict the demand for firefighters in urban fires was divided into two parts. The overall process is illustrated in Fig. 1.

Part I was data preprocessing. In **step1**, data preprocessing consisted of data cleaning, data merging and anomaly detection. Data cleaning process aimed to eliminate the effects of poor-quality data. During the data merging process, fire data was combined with weather data to create a fused dataset. Following data cleaning and merging, four different anomaly detection methods were used to identify anomalies in the processed data. Datasets 1, 2, 3 and 4 represent the results of anomaly detection performed by the MCD, ABOD, Auto Encoder and Iforest algorithms, respectively.

Part II focused on the construction of urban firefighters demand prediction model. In **step2**, during the construction of feature subsets, four different feature selection methods (XGBoost, GBDT, Random

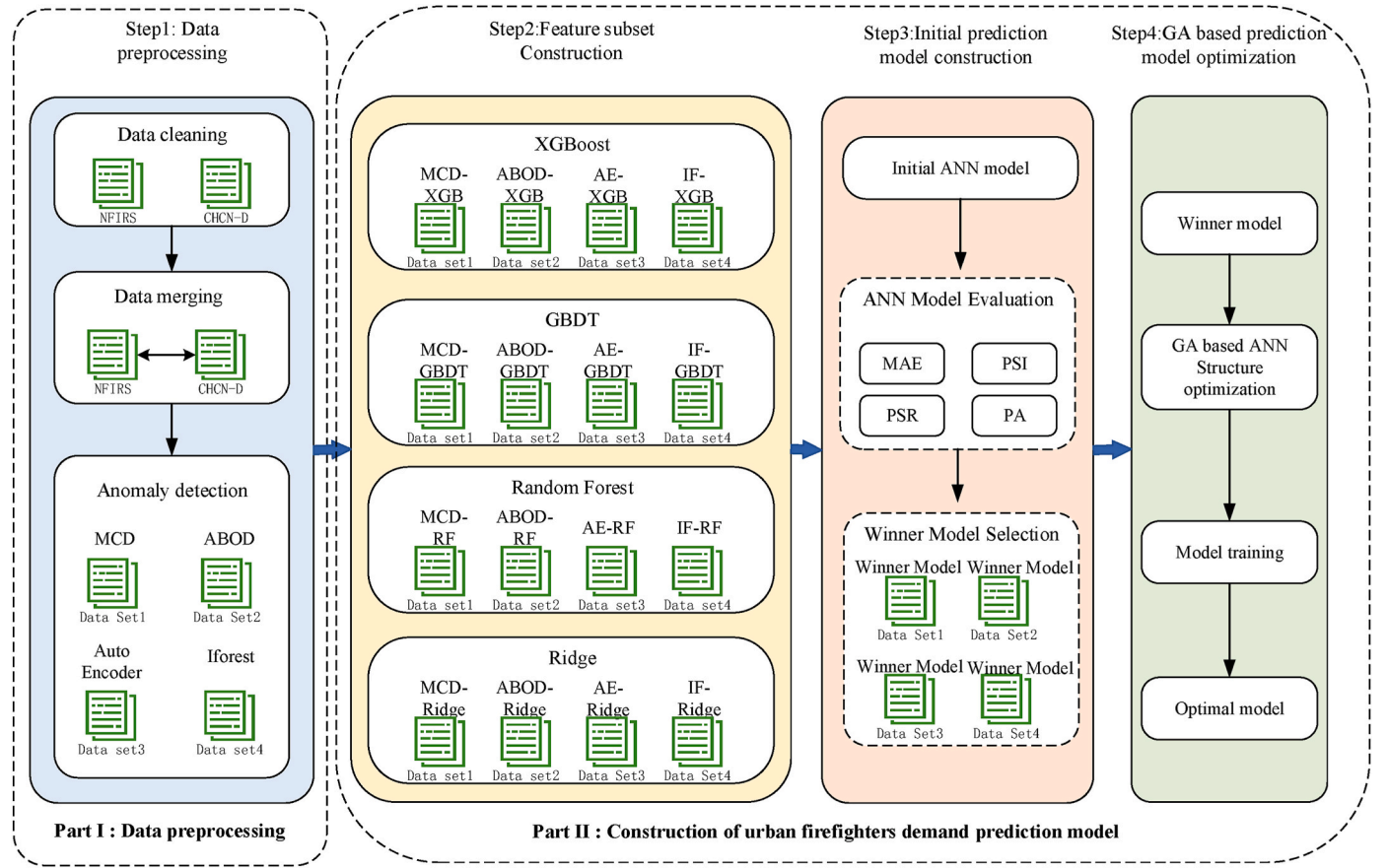


Fig. 1. Proposed method for predicting the demand for urban firefighters.

Forest and Ridge) were respectively combined with the four datasets obtained from anomaly detection. This resulted in a pairwise combination of 16 datasets. After that, **step3** involved putting each of the above dataset into the initial demand prediction model and evaluate its effectiveness. The initial demand prediction model refers to a single hidden-layer back-propagation artificial neural network with an unoptimized structure. The 16 datasets obtained in step2 were fed into it for operation, and the results were assessed with four types of indicators: Mean Absolute Error (MAE), Proportion of Severe Inadequate (PSI), Proportion of Acceptable (PA) and Proportion of Severe Redundancy (PSR). By ranking and assigning scores to the result values, the best feature selection method under each anomaly detection method was finally identified. The four winning initial demand prediction models were taken as the next optimization targets. Subsequently, in the final **step4**, winner models were further enhanced through structure optimization using the genetic algorithm (GA). GA achieved optimization by heuristically searching for the structure of ANN in terms of the number of hidden layers, the number of neurons per layer and activation function per layer. Then, the optimized structure of ANN was obtained after training and comparing various models.

3.1. Data preprocessing

Unlike previous studies, this paper aimed to propose a novel method based on historical data to predict the demand for firefighters. Recognizing the crucial impact of data quality on model effectiveness, substantial effort was dedicated to the data preprocessing phase. This part could be divided into three processes: data cleaning, data merging and anomaly detection.

During data cleaning, missing values, mistyped values, extreme values and low-frequency values were removed. Data merging process

linked the fire data from the NFIRS database with the weather data from the GHCN-D dataset by date and station information. The anomaly detection process used four main stream algorithms: MCD, ABOD, Auto Encoder and Iforest. Each of them belongs to different categories.

Fig. 2 provides a detailed overview of the specific preprocessing

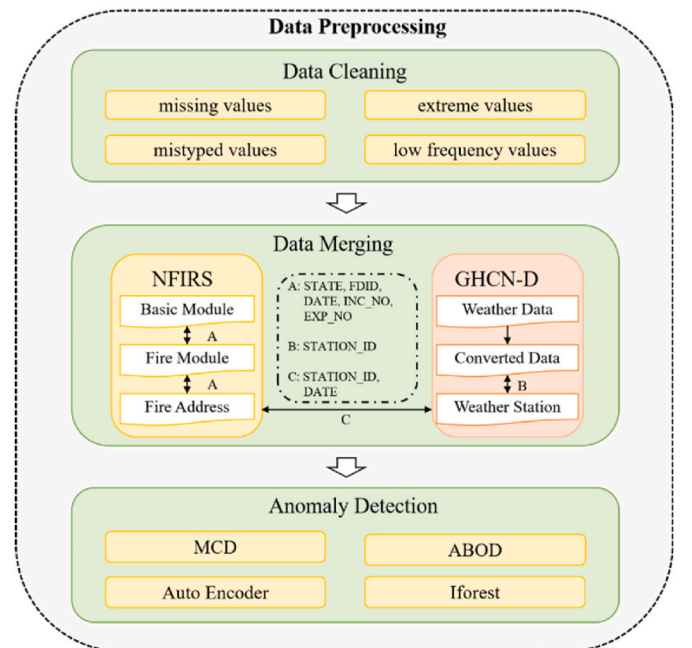


Fig. 2. Data preprocessing process.

process, which will be further described below.

3.1.1. Data sources

NFIRS (National Fire Incident Reporting System) is a modular incident reporting system that has been designed and developed by the U.S. Fire Administration (USFA) with the aim of providing data to support fire assessment and research. With over 15,000 fire departments in more than 40 major metropolitan areas, NFIRS stands as the largest fire incident reporting system globally, covering approximately 70 % of all fire incidents in the United States. The latest version, NFIRS 5.0 encompasses all incidents where fire departments have been involved in responding to fires. To ensure comprehensive and organized record of incident information, the system consists of eleven information modules, and each one has a specific purpose. The description of each module is shown in Table 1.

Basic Module requires the inclusion of basic fire incident information and address-related details for all types of incidents. The remaining modules are completed based on the specific requirements of each incident. NFIRS database offers structured data, where each line in the data file represents a fire incident. The incidents are uniquely identified by a composite primary key consisting of “State”, “Fire Department Identification”, “Incident Date”, “Incident Number” and “Exposure Number”. “Exposure Number” indicates the sequential order in which buildings are affected by the fire. It is used to describe situations where the fire spreads from the initial fire-starting building to another building with fire protection measures in place. The number assigned to the initial fire-starting building is “000”.

The weather data used in this paper were obtained from the Global Historical Climatology Network Daily (GHCN-D), the dataset provided by the National Oceanic and Atmospheric Administration (NOAA). GHCN-D consists of daily climate observations from land-based stations worldwide. It includes various measurements such as maximum

temperature, minimum temperature, precipitation, snowfall and snow depth. The data files are organized by year, with each station providing one row a day for each indicator. Table 2 presents a sample of raw weather data. The attributes listed under the “ELEMENT” column in the table represent specific meteorological or climatic measurements or statistics. For instance, PRCP corresponds to precipitation, TMIN represents minimum temperature, SNOW indicates snowfall, and so on. “DATA_VALUE” represents the observed value. “M_FLAG” represents the measurement flag, where the value T indicates a trace of precipitation, snowfall or snow depth. “Q_FLAG” represents the quality flag, with value L indicating a failed check on the length of the multiday period and value I indicating a failed internal consistency check. “S_FLAG” represents the source flag, where value 7 corresponds to the U.S. Cooperative Summary of the Day, C means data from Environment Canada, and W indicates the WBAN/ASOS Summary of the Day from NCDC’s Integrated Surface Data. “OBS_TIME” represents the time of observation, which is 24-h clock time (in a 2-digit hour, 2-digit minute format) of the observation recorded in local time at the station.

3.1.2. Data cleaning and data merging

This paper focuses on predicting the demand for urban firefighters. For this purpose, the following part built on existing research by selecting features relevant to firefighters dispatch to construct the dataset.

Firstly, the data of Fire Module and Basic Module in NFIRS were cleaned by deleting data with null values. This process resulted in 16962 Fire Module records and 348783 Basic Module records. It should be noted that the Basic Module contains various types of fire, such as urban fire, forest fire, chemical plant fire, etc. This paper is based on the Fire Module, and the Basic Module supplements the information of urban fires. After completing the above step, the data cleaning continued for both continuous and discrete features. Based on the results of data exploration, the following rules were developed for continuous features in order to remove irregular data rows.

- (1) Data rows where both the property value (PROP_VAL) and the content value (CONT_VAL) are 0, the building length (BLDG_LGTH) or the building width (BLDG_WIDTH) is 0, and suppression personnel (SUP_PER) is 0.
- (2) Data rows where the number of dwelling units (NUM_UNIT) or the number of buildings involved (BLDG_INVOL) is less than 0.
- (3) Data rows where the floor of fire origin (FIRE_ORIG) is higher than the floor height (BLDG_ABOVE) or lower than the number of basement floors (BLDG_BELOW).
- (4) Data rows where the interval between arrival time (ARRIVAL) and alarm time (ALARM) exceeds 24 h.

For discrete features, in order to reduce the impact of fire incident records with feature values that occur at low frequencies, the following deletion rules were formulated to delete the records with special values.

- (1) Incident type (INC_TYPE) or property use (PROP_USE) occurs less than 50 times.
- (2) Structure type (STRUC_TYPE), structure status (STRUC_STAT) or first response action (ACT_TAK1) occurs less than 30 times.

It should be noted in particular that in some non-residential fires, such as unconventional fires containing poisons and explosives, the pre-incident plan specifies the minimum number of firefighters demanded. While in this paper, the Property Use (PROP_USE) feature of non-residential fires is aimed at conventional types. It means that when a non-residential fire occurs, there are no toxic, explosive and other potentially dangerous risk factors.

According to the format and distribution of GHCN-D data, several preprocessing operations were required before integrating with NFIRS fire incident data. These operations are outlined as follows.

Table 1
Introduction of modules in NFIRS.

Module name	Module Description
Basic Module	It is utilized to provide general information regarding each incident that the fire department responds to.
Fire Module	It is utilized to describe each fire incident handled by the fire department.
Structure Fire Module	It is utilized to describe each structure fire that the fire department responds to, and this module is used in conjunction with the Fire Module.
Civilian Fire Casualty Module	It is utilized to report injuries or fatalities of civilians or other emergency personnel (such as police officers, non-fire department or EMS personnel) that are associated with a fire incident.
Fire Service Casualty Module	It is utilized to report injuries and fatalities related to firefighters.
EMS Module	It is utilized to report all medical incidents in which the department provided the primary patient care.
Hazardous Materials Module	It is utilized to report incidents involving spills or releases of 55 gallons or more of hazardous materials or special hazardous materials actions were taken. It can be used in conjunction with other modules, such as the Fire Module, to provide comprehensive details regarding hazardous materials incidents.
Wildland Fire Module	It is filled out to report incidents specifically related to wildland or vegetation fires. This module is used instead of the Fire Module when documenting wildland fire incidents.
Apparatus or Resources Module	It is utilized by fire departments to report data specific to each apparatus that responds to an incident. This module is not used if the Personnel Module is used.
Personnel Module	It is utilized by fire departments to report the same information as the Apparatus or Resources Module. Additionally, it provides the capability to track the personnel associated with each apparatus.
Arson Module	It is utilized to provide additional information on fires that have been classified by the department as “intentionally set”.

Table 2

Samples of original weather data.

ID	DATE	ELEMENT	DATA_VALUE	M_FLAG	Q_FLAG	S_FLAG	OBS_TIME
USC00054054	20160215	PRCP	0	T	L	7	0700
US009052008	20160106	TMIN	−43	−	−	C	−
USW00024032	20161212	SNOW	0	T	I	W	−

- (1) Selected all U.S. weather station data starting with “US” using regular matching.
- (2) Converted the granularity of the raw data into the form of a weather station ID plus a date, where each row represents the daily observation data for a specific weather station.
- (3) Removed data from weather stations with incomplete main observation indexes (e.g., average wind speed, precipitation, snowfall, snow depth, maximum air temperature and minimum air temperature).
- (4) Removed data from weather stations with an incomplete number of recorded days in 2016.
- (5) After completing the above steps, it was discovered that there were no weather stations with complete observations of meteorological indicators in Mississippi. Therefore, separate processing is required. Based on the prioritization of data quality requirements, data related to weather stations with insufficient days of records and missing temperature-related records were deleted first. Among the remaining weather stations, incomplete data was mainly due to missing wind speed or snowfall data. Considering the climate in Mississippi, which is located in the southern part of the United States with minimal snow throughout the year, weather stations with wind speed data were retained, and the snowfall attribute was assigned a value of zero.
- (6) Merged the Mississippi data with weather data from other states to obtain a complete weather dataset.

After data cleaning, in data merging process, the composite primary key was applied to connect Fire Module with Basic Module to get the complete NFIRS fire incident data. In the integration of NFIRS fire incident data with external weather data, this part utilized Bing Maps API to obtain the latitude and longitude of the city where each fire incident was located. Such city location data was recorded in NFIRS fire address dataset (incidentaddress.txt). These coordinates were then matched with the latitude and longitude of each weather station in weather station data file of the GHCN-D dataset (ghcnd-stations.txt). As a result, the nearest weather station to each fire incident was identified. Finally, the weather data for the corresponding date was obtained from the matched weather station ID.

Based on the aforementioned preprocessing steps, Table 3 presents the source, name and value range of each feature in the fused fire incident dataset. In terms of features availability, the urban fire information can be obtained through the description of alarm person (F1–F12, F16, F17). Through the real-time organization of the fire department, information on fire response is available (F13–F15). For weather information, it can be obtained through monitoring by the meteorological services (F18–F23).

Additionally, to ensure that the features in the fused dataset provide valuable information for the demand prediction model, this section employed feature extraction to reconstruct certain features that may lack practical significance, such as incident date and alarm time. The reconstructed features aim to eliminate correlations among original features while enhancing their significance. The original features and their corresponding generated features are listed in Table 4.

Finally, the fused fire incident dataset consisted of 3521 complete urban fire incident records, with each record containing 21 features (F1 through F28 minus the 7 primary features that were replaced by the generated features).

Table 3

Description of primary features.

Data Source	No.	Feature name	Values
NFIRS Fire Incidents	F1	Incident Date (INC_DATE)	E.g.: 01012016
	F2	Number of Residential Units (NUM_UNIT)	[0, 500]
	F3	Number of Buildings Involved (BLDG_INVOL)	[0, 100]
	F4	Structure Type (STRUC_TYPE)	Enclosed building (1); Portable/Mobile structure (2); Other type of structure (0)
	F5	Structure Status (STRUC_STAT)	Under construction (1); In normal use (2); Idle (3); Not routinely used (4); Under major renovation (5); Vacant and secured (6)
	F6	Building Height: Stories Above Grade (BLDG_ABOVE)	[0, 60]
	F7	Building Height: Stories Below Grade (BLDG_BELOW)	[0, 11]
	F8	Building Length (BLDG_LGTH)	[0, 150150]
	F9	Building Width (BLDG_WIDTH)	[0, 5120]
	F10	Fire Origin (FIRE_ORIG)	[-1, 10]
	F11	Incident Type (INC_TYPE)	Building fire (111); Fire in structure (not in a building, 112); Cooking fire (113); Fire in a mobile home (121); Fire in a motor home (122)
NFIRS Basic Incidents	F12	Property Use (PROP_USE)	Restaurant or cafeteria (161); 1- or 2-family dwelling (419); Multifamily dwelling (429); Residential, other (400); Manufacturing (700); Outbuilding or shed (808); Parking garage (881); Extinguishment by fire service personnel (11); Salvage and overhaul (12); Fire control or extinguishment, other (10); Ventilate (51); Incident command (81); Investigate (87)
	F13	Primary Action Taken (ACT_TAK1)	E.g.: 010120161208
	F14	Alarm Date and Time (ALARM)	E.g.: 010120161228
	F15	Arrival Date and Time (ARRIVAL)	E.g.: 010120161228
	F16	Property Value (PROP_VAL)	[0, 895020000]
	F17	Contents Value (CONT_VAL)	[0, 500000000]
	F18	Average Wind (AWND)	[0, 127]
	F19	Precipitation (PRCP)	[0, 1976]
	F20	Snowfall (SNOW)	[0, 671]
	F21	Snow Depth (SNWD)	[0, 710]
	F22	MAX Temperature (TMAX)	[-28.2, 43.3]
GHCN-D Database	F23	MIN Temperature (TMIN)	[-33.8, 28.9]

Table 4
Generated features from primary features.

Primary Features	No.	Generated features	Values
F1	F24	SEASON	Spring (0), Summer (1), Autumn (2), Winter (3)
F14, F15	F25	Response Time (RES_TIME)	[0,131]
F14	F26	PERIOD	0–6 a.m. (0), 7–12 a.m. (1), 1–6 p.m. (2), 6–12 p.m. (3)
F8, F9	F27	Building Area (BLDG_AREA)	[0, 1200000]
F22, F23	F28	Average Temperature (TAVG)	[-31, 35.8]

3.1.3. Anomaly detection

The previous data cleaning process eliminated extreme values on a single feature. However, current dataset still contained some extreme records. Although the inputs were similar, these records showed significant output gaps compared to others. For example, Table 5 displays two records with identical values on the unlisted features and similar values on the listed features, yet their outputs (SUP_PER) differ by 19. Upon analysis, on the one hand, data reporting errors could lead to record conflicts. On the other hand, only the features relevant to firefighters dispatch were selected to construct the dataset in this paper. And it is possible that the values of unselected features could be very different, resulting in these conflicting records. To reduce the impact of such data records on training and prediction, this paper used anomaly detection methods to identify abnormal samples.

Given that the existing dataset lacked labels to indicate outliers, an unsupervised algorithm was necessary to address the problem. Following the core principles of unsupervised anomaly detection, machine learning algorithms can be classified into four categories: proximity-based algorithms, linear algorithms, neural network algorithms and outlier ensemble algorithms. The popular algorithms of them are listed in Table 6.

Firstly, proximity-based algorithms focus on the distribution difference between abnormal data samples and normal data samples, which can be identified in terms of distance, density or angle. Among them, the Angle-Based Outlier Detection (ABOD) is more stable in high-dimensional space [56]. Secondly, algorithms based on linear assume that there is a linear correlation between the multidimensional data. Principal Component Analysis (PCA) is a classic algorithm in this category [57]. When mapping the data into lower dimensions, the outliers tend to have higher errors to be recognized. Similarly, Minimum Covariance Determinant (MCD) calculates the weighted Euclidean distance of each data sample to the hyperspace, which is formed by reconstructed independent features. Compared to PCA, MCD is more robust [58]. Thirdly, there are two types of algorithms based on neural network: one is the Auto Encoder and the other is the generative adversarial network-based algorithm. Auto Encoder is similar to PCA when detecting outliers. The difference is that Auto Encoder uses neural networks for data compression, and the data samples with high errors after decompressing are considered abnormal [59]. Single-objective and multi-objective generative adversarial active learning (SO_GAAL, MO_GAAL) use generators to generate more abnormal data samples and then use discriminators to distinguish the normal ones and abnormal ones from ordinary datasets [60]. Outlier ensemble algorithms combine detectors on subsets of different dimensions into an integrated detector.

Table 5
Samples of abnormal records.

PROP_VAL	BLDG_AREA	ARR_TIME	SEASON	AWND	TAVG	SUP_PER
40000	1000	6	1	27	14.2	20
43500	1360	3	2	30	25	1

Table 6
Different algorithms of anomaly detection.

Category	Algorithm
Proximity-Based	KNN, LOF, CBLOF, ABOD
Linear	OCSVM, PCA, MCD
Neural Network	Auto Encoder, SO_GAAL, MO_GAAL
Outlier Ensemble	Feature Bagging, Isolation Forest

This kind of algorithm has stronger generalization ability, including Feature Bagging [61] and Isolation Forest [62].

According to previous studies, each type of algorithm has its own advantages, and their performance varies across different domains and datasets. Therefore, the proposed demand prediction method aimed to find an optimal algorithm according to the characteristics of the dataset in experiments. In this paper, the dataset was unlabeled and mixed with abnormal data samples. Therefore, algorithms with high accuracy and robustness should be prioritized, even if it comes at the cost of efficiency. Taking these considerations into account, MCD, ABOD, Auto Encoder (AE) and Isolation Forest (IF) algorithms were selected as representatives of each category.

Since many machine learning algorithms assume that the data conforms to a normal distribution, data distribution of continuous features was explored to ensure the prediction effect. According to the skewness and kurtosis of them, all continuous features except for “FIRE_ORIG” and “TAVG” had severe skewed distribution. Therefore, they were subjected to a Box-Cox transformation. Set the value of the i th record on the k th feature is x_{ik} , then the transformed value x_{ik}^* can be expressed as shown in Equation (1):

$$x_{ik}^* = \log(x_{ik} + 1) \quad (1)$$

The transformation made the primary data closer to a normal distribution and reduced the dimensional differences between continuous features. After that, values of continuous features were mapped to the 0–1 interval according to the maximum and minimum values. Set max value of the k th feature is x_{kmax} , min value of the k th feature is x_{kmin} , the transformed value x_{ik}^* can be expressed as shown in Equation (2):

$$x_{ik}^* = \frac{x_{ik} - x_{kmin}}{x_{kmax} - x_{kmin}} \quad (2)$$

Additionally, to ensure that the dataset could be directly utilized for constructing demand prediction models after anomaly detection, one-hot encoding was performed for discrete features. After the above steps, the dataset was processed using MCD, ABOD, AE and IF algorithms respectively. As a result of the anomaly detection, four cleaned datasets were generated.

3.2. Construction of GA-ANN demand prediction model

In this section, a multi-optimization (from the perspective of parameter optimization and structural optimization) demand prediction model was proposed for predicting the number of firefighters demanded. The modeling process involved four main steps: feature subset construction, initial demand prediction model construction, model evaluation and model optimization.

During the feature subset construction process, the independence of the data was tested using Spearman’s coefficient. Subsequently, four feature selection methods (XGBoost, GBDT, RF and Ridge) were employed to select the most relevant features respectively. Four datasets

generated by different feature selection methods were then utilized to train initial demand prediction models. A series of rules were designed to evaluate the performance of them, including MAE, PSI, PA and PSR. In order to optimize the structure of initial demand prediction model, this part proposed an optimization process based on genetic algorithm (GA). After ten-fold cross validation, the optimized GA-ANN demand prediction model was finally obtained. The complete process is illustrated in Fig. 3.

3.2.1. Feature subset construction

Feature selection methods can be categorized into three types based on different evaluation strategies: filter, wrapper and embedded algorithms [63]. Filter algorithms evaluate each feature individually, using metrics such as chi-squared test scores, information gain or correlation coefficients. Features are then selected according to a predefined threshold. These algorithms are highly versatile and efficient [64], but their accuracy is not always as good as the other two types. Unlike filters, wrapper algorithms directly use the performance of the final demand prediction model as the evaluation criteria for feature subsets. Recursive feature elimination (RFE) is a classical wrapper algorithm. It fits the demand prediction model and removes the weakest feature at each iteration until the specified number of features is reached. Since wrappers directly select features for the final demand prediction model, their effectiveness is generally superior to others. However, as the number of features increases, the computational cost can rise exponentially. This is due to the need to repeat model training for each feature subset. Embedded algorithms are integrated within algorithms that have their own feature selection methods, such as tree-based algorithms and regularized linear regression algorithms. Like wrappers, embedded algorithms also consider the interaction of features, but they are faster in a similar way to filters.

The first step of this part employed the correlation coefficient selection algorithm, which is a filter algorithm to examine the correlation of features. The Spearman's rank correlation coefficient denoted by ρ , which is commonly used to reflect the degree of correlation between two features and can be written as shown in Equation (3):

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \quad (3)$$

X and Y denote two different features, and X_i and Y_i represent the i th X and the i th Y respectively. In this equation, d_i is the level difference between X_i and Y_i , and n represents the number of records. It is clear that

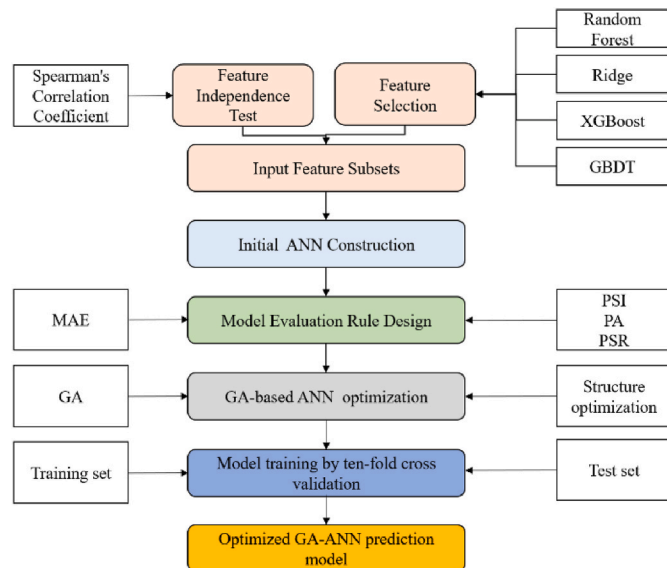


Fig. 3. Process of construction of GA-ANN demand prediction model.

the range of ρ is $[-1, 1]$. If the absolute value of ρ is close to 1, then the two features have stronger correlation. According to the previous studies, ρ should be less than 0.6 to ensure that the features are not highly correlated, otherwise it could lead to multicollinearity problem.

Since the artificial neural network is a black-box method and requires a long training time for each iteration, it is not suitable as the base model for a wrapper algorithm. Therefore, it's appropriate to use an embedded algorithm for the construction of feature subsets. In order to find the optimal feature selection method, the proposed method employed four embedded algorithms, including eXtreme Gradient Boosting (XGBoost), Gradient Boosting Decision Tree (GBDT), Random Forest (RF) and Ridge Regression. The above algorithms were used to individually identify feature subsets related to target value. Initial demand prediction models were then constructed, and feature subsets were evaluated based on the performance of them. The input feature subset of the final demand prediction model was determined based on the results of initial demand prediction models.

3.2.2. Initial demand prediction model construction

Modeling the number of demand firefighters is a complex nonlinear problem. This process requires a model with strong self-learning capability that can effectively handle both quantitative and qualitative features. Compared with other machine learning models, ANN has a strong ability to fit nonlinear problems. At the same time, its excellent parallel computing ability ensures that it can be efficiently applied and iterated on massive datasets. Therefore, the initial demand prediction model based on ANN was constructed.

ANN is a mathematical model inspired by biological neural network. It is designed to implement complex mappings between inputs and outputs, enabling it to perform specific tasks much like the human brain. An ANN consists of an interconnected sequence of artificial neurons, typically arranged in layers or vectors. The connection of neurons between adjacent layers is fully connected, which means every neuron in this layer connects to all neurons in the previous one.

As depicted in Fig. 4, the layer that receives external data is input layer, and the layer that produces the ultimate result is output layer. Given that the firefighters demand prediction is a regression problem, there is only one neuron in output layer. Between the input and output layers, there can be one or more hidden layers.

In order to fairly compare the performance of the initial demand prediction models on different feature subsets, this part considered adopting the same strategy to set the structure of the ANN.

First, the number of hidden layers and the number of neurons in each layer were determined. The initial demand prediction model consisted

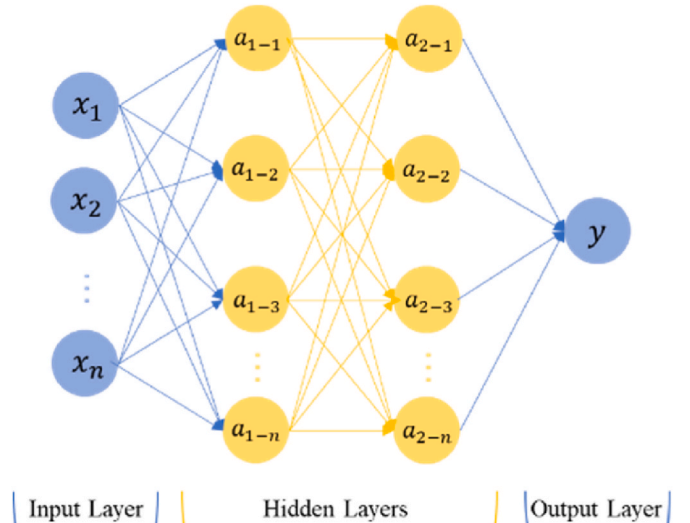


Fig. 4. Illustration of artificial neural network architecture.

of an input layer, a single hidden layer and an output layer. Referring to Kolmogorov's theorem, the number of neurons in the input layer was set equal to the dimension of the input data, denoted as K . The number of neurons in the hidden layer was set to $K * 2 + 1$, and the output layer contained only one neuron.

Next, the activation function was required to be determined. The activation functions can be categorized into linear and nonlinear. The nonlinear activation functions introduce a nonlinear factor into the neuron, making the artificial neural network approximate any nonlinear function. The most widely used activation function is the Sigmoid, which is easy to derive and map the value into the interval between 0 and 1. The hard_Sigmoid is an approximation to Sigmoid, which is faster to compute. However, as the number of hidden layers increases, the gradient is prone to disappearing. The hyperbolic tangent function (Tanh) has a faster convergence rate, but it also suffers from the problem of vanishing gradients. The rectified linear unit (ReLU) converges quickly in stochastic gradient descent (SGD) and solves the vanishing gradient problem. But when encountering negative data, ReLU converts it to 0, which can impact the performance of the model. The equations for the activation functions are given in Table 7. Since there is no perfect activation function, the firefighters demand prediction model in this paper selected different activation functions for different layers, and the optimized solution was determined through multiple experiments. As the initial demand prediction model was single hidden layer neural network, Sigmoid was selected as the activation function of the hidden layer through experimental comparison.

In this paper, the demand prediction model was a feed-forward artificial neural network with back propagation (BP) learning process. The BP algorithm is adept at fitting non-linear function [65], and can effectively deal with the non-linear relationship between characteristic parameters and the number of firefighters. Assuming there are n inputs and one output in the network, hidden layer has s neurons. From input layer to hidden layer, threshold of the j th neuron in hidden layer is represented by b_j . The weight of the connection between the i th neuron (represented as x_i) in input layer and the j th neuron in hidden layer is represented by w_{ij} . The activation function of the hidden layer is f_1 . Then the input and output of j th neuron in hidden layer can be expressed as shown in Equation (4) and Equation (5):

$$Hidden_{input_j} = \sum_{i=1}^n w_{ij} x_i \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, s) \quad (4)$$

$$Hidden_{output_j} = f_1(Hidden_{input_j} - b_j) \quad (5)$$

From hidden layer to output layer, weight of each link is w_j , threshold value is b , and the activation function is f_2 . Then output $y_{predict}$ of the model can be expressed as shown in Equation (6):

$$y_{predict} = f_2\left(\sum_{j=1}^s w_j Hidden_{output_j} - b\right) \quad (j = 1, 2, \dots, s) \quad (6)$$

Suppose true value of the i th sample is $y_{true,i}$, predictive value of the i th sample is $y_{predict,i}$, N is the size of the data. The loss function can be expressed as shown in Equation (7):

$$loss = \frac{1}{N} \sum_{i=1}^N (y_{predict,i} - y_{true,i})^2 \quad (7)$$

ANN back-propagates the output loss to hidden layer until it propagates to the input layer. During the process of back propagation, the values of various parameters are adjusted according to loss. Above process is iterated until convergence. The BP algorithm is based on the gradient descent strategy, and there are various optimizers available for parameter adjustment. The common optimization algorithms are Gradient Descent (GD), Momentum and Adaptive gradient algorithm (Adagrad). GD algorithm is easy to fall into local optimization, but Momentum algorithm can improve this problem [66]. Adagrad algorithm is more effective in improving the learning rate. Based on this, the paper adopted the NAdam algorithm, which has the advantages of both Momentum algorithm and Adagrad algorithm, to optimize the parameters of the initial demand prediction model according to the prediction loss.

The aforementioned settings ensured consistency in the structure of all initial demand prediction models, which allowed for more accurate comparisons of their performance on different subsets of features.

3.2.3. Model evaluation

A comprehensive and robust system of model evaluation indicators is important to ensure that the demand prediction model achieves optimal performance. The purpose of this paper is to predict the proper number of firefighters at the alarm time, and the exact number is needed as a reference basis. For this purpose, this paper chose the Mean Absolute Error (MAE) as a prediction indicator. MAE is a commonly used evaluation indicator for regression models, which can be calculated as shown in Equation (8):

$$MAE = \frac{1}{N} \sum_{i=1}^N |observed_i - predicted_i| \quad (8)$$

where N represents the size of the dataset, $observed_i$ represents the real target value of the i th record, and $predicted_i$ denotes the predicted value of it.

In order to enable a systematic evaluation of the predictive effectiveness of demand prediction model, it's important to not only consider the overall error, but also evaluate the error for each individual fire incident. As containers to carry important fire equipment and transport of firefighters, fire trucks are an indispensable part of the fire dispatch process. Therefore, based on the load of fire trucks, this paper evaluated each individual fire incident error of the number of firefighters prediction. Considering that a fire truck is typically staffed with 5 firefighters in general, this paper defined three indicators based on the error between predicted value and actual value using a criterion of 5 firefighters. These indicators are overall relative errors applicable to the firefighter demand scenario in this paper. These indicators are: Proportion of Severe Inadequacy (PSI), Proportion of Acceptability (PA) and Proportion of Severe Redundancy (PSR).

Severe Inadequate (SI) is used to determine if the actual number of firefighters dispatched to an incident exceeds the predicted value by more than five personnel. Proportion of Severe Inadequacy (PSI) stands for the proportion of incidents where the number of firefighters dispatched was severely underpredicted. SI and PSI can be expressed as shown in Equation (9):

$$SI = \begin{cases} 1, & (observed_i - predicted_i) > 5 \\ 0, & otherwise \end{cases}, PSI = \frac{SI}{N} \quad (9)$$

Acceptability (A) is used to determine if the error between the predicted and actual number of dispatched firefighters is within five personnel. Proportion of Acceptability (PA) represents the proportion of predicted number of firefighters that are substantially equivalent to actual values. A and PA can be expressed as shown in Equation (10):

Table 7

The equations for the activation functions.

Activation Function	Equation
Linear	$f_{Linear}(x) = x$
Sigmoid	$f_{Sigmoid}(x) = \frac{1}{1 + e^{-x}}$
hard_Sigmoid	$f_{hard_Sigmoid}(x) = \max\left(0, \min\left(1, \frac{2x+5}{10}\right)\right)$
Tanh	$f_{Tanh}(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
ReLU	$f_{ReLU}(x) = \max(0, x)$

$$A = \begin{cases} 1, & |observed_t - predicted_t| \leq 5 \\ 0, & otherwise \end{cases}, PA = \frac{A}{N} \quad (10)$$

Severe Redundancy (SR) is used to determine if the predicted number of dispatched firefighters exceeds the actual value by more than five personnel. Proportion of Severe Redundancy (PSR) represents the proportion of incidents where the number of dispatched firefighters was overpredicted. SR and PSR can be expressed as shown in Equation (11):

$$SR = \begin{cases} 1, & (predicted_t - observed_t) > 5 \\ 0, & otherwise \end{cases}, PSR = \frac{SR}{N} \quad (11)$$

SI means that there is a large demand for additional dispatch, so PSI should be as small as possible. Meanwhile, PSR reflects the wastage of resources, which should also be minimized. Correspondingly, the PA should be as large as possible, indicating a higher proportion of incidents with the appropriate number of firefighters dispatched.

To validate the effectiveness of the demand prediction model, ten-fold cross validation was employed during training. The dataset was evenly divided into 10 parts, taking turns to make nine of them for training and one for validation, and the average of ten results was used as the final result.

From step3 of Fig. 1, it can be seen that initial demand prediction models were trained on different input feature subsets. A scoring rule was designed to evaluate them based on their scores of the four indicators (MAE, PSI, PA and PSR). This rule ranked the initial demand prediction models according to the values of each indicator and assigned a score from 1 to 4 for each dimension, indicating poor to good performance. For each of them, its final score was the sum of scores for four indicators. A higher score indicated better predictive performance for the initial demand prediction model. In cases where two models had the same final score, the one with the lower MAE was considered the optimal choice.

3.2.4. Model structure optimization

According to the model evaluation rules from Section 3.2.3, the best-performing ones among all initial demand prediction models was identified. However, their structures were relatively simple and required further optimization. The performance of an artificial neural network model depends heavily on its structure. An ANN with simple structure may have poor learning ability, making it hard to better fit nonlinear problems, while overfitting may present when the ANN has complex structure. Grid search is a popular method for structure optimization, but it can be time-consuming as the number of layers increases [67]. Therefore, many studies set the structure empirically according to certain criteria, such as the minimization of a quadratic cost [68]. However, these methods are highly subjective, and it is challenging to realize optimal performance of the model. In terms of all these factors, this section proposed the GA-ANN demand prediction model, which used genetic algorithm to optimize model structure.

Genetic Algorithm (GA) is one of the most popular evolutionary algorithms [69], which uses techniques from evolutionary biology such as inheritance and mutation [70]. The GA-ANN model encoded a possible ANN structure as an individual and realized structure optimization through population evolution. In the GA optimization process, coding rules were defined first. After population initialization, fitness of each individual was calculated, tournament strategy was used in selection phase for selecting the optimal individuals to generate offspring. Then crossover and mutation operations were performed to obtain the optimal individual (the optimal demand prediction model structure) through continuous iteration.

ANN structure includes the number of layers, the number of neurons in each layer and the type of activation function. Mutation is one of the key implementations of population evolution. In this context, the mutation of individuals was divided into three types: add layers, delete layers and mutations within a layer. Mutations within a layer included changing the number of neurons, changing the activation function or

changing both of them. When changing the number of neurons, there are options to add one, delete one or reinitialize. The pseudo code of structure optimization process of GA-ANN model is shown in Table 8.

4. Results and discussion

According to the urban fire firefighter demand prediction method proposed in this paper, four datasets created by the four anomaly detection methods were obtained respectively after data preprocessing in Section 3.1. In the following, results of the urban fire firefighters demand prediction model construction will be analyzed and discussed using these 4 datasets. The construction results of feature subsets, selection results of initial demand prediction models, optimization results of GA-ANN model and comparative experiment results are shown in this section.

Table 8

Pseudo code of structure optimization process of GA-ANN model.

GA optimization process started
Design coding rules
<i>Individual coding design</i>
Define max hidden layers (MAX_LAYERS)
Define min neurons in each hidden layer (MIN_NEURONS_LAYER)
Define max neurons in each hidden layer (MAX_NEURONS_LAYER)
Define the collection of activation functions (ACTIVATIONS)
<i>Population evolution design</i>
Define population size (POP)
Define generations (GEN)
Define the number of individuals selected in Tournament (tour_SIZE)
<i>Probability of mutations</i>
Define probability of mutations (P):
mutation within a layer (P₁), add layers (P₂), delete layers (P₃),
change the number of neurons (P₁₁), activation function (P₁₂), both (P₁₃)
add one (P₁₁₁), delete one (P₁₁₂), reinitialize (P₁₁₃)
<i>Parameter optimizer</i>
NAdam parameter optimizer for weights and thresholds of hidden layers neurons: (opt_WEIGHT), (opt_THRESHOLD)
Population initialization
Initial the number of hidden layers (NUM_LAYERS):
Randomly from [1 , MAX_LAYERS]
Initial the number of neurons each layer (NUM_NEURONS_LAYER):
Randomly from [MIN_NEURONS , MAX_NEURONS]
Initial activation function of each layer: Randomly from ACTIVATIONS
For i = 1 to GEN do
Calculate fitness
For n = 1 to 10 do # Ten-fold cross validation
Run Parameter optimizer:
current_weight ← optimized_WEIGHT
current_threshold ← optimized_THRESHOLD
Calculate $e_n = \text{current MAE}$
End for n
Calculate the average error for each training set $e = \frac{1}{10} \sum_{n=1}^{10} e_n$
Return fitness $s = f(e)$
Selection operation
Initial the number of OFFSPRING = 0 (OFF = 0)
<i>Tournament Strategy</i>
WHILE OFF < POP
Select tour_SIZE individuals randomly
Select the highest fitness individual into the OFFSPRING
OFF = OFF + 1
Return OFFSPRING as new population
Crossover
Select two individuals in OFFSPRING randomly for single-point
Mutation
Perform probability of mutations (P) on OFFSPRING
End for i
Return the Optimized ANN Structure:
optimized_NUM_LAYERS
optimized_NUM_NEURONS_LAYER
optimized_ACTIVATIONS
End GA optimization process

4.1. Results of feature selection

According to Equation (3), the correlation coefficient matrix of features in the primary dataset is plotted as a heatmap, as shown in Fig. 5. If the color of the block is closer to dark blue, it indicates a stronger positive correlation between the two features. Conversely, if the color is closer to light yellow, it indicates a stronger negative correlation.

The results show that the absolute values of the Spearman's coefficient between all features are less than 0.6, which means all of them can be preserved from the perspective of feature independence. After that, this section uses a combination of anomaly detection methods and feature selection methods as the name of each dataset. The feature selection results on the four datasets based on XGBoost, GBDT, Random Forest and Ridge are shown in Fig. 6. The horizontal axis represents the input features. Since the discrete features have been one-hot encoded, each possible value is uniquely identified in the dataset. For example, "INC_TYPE_111" represents the incident type of "building fire". The vertical axis represents the datasets. Finally, the dark blue grid represents that the feature is selected in the corresponding dataset.

According to Fig. 6, certain values in the discrete features, such as "INC_TYPE_112" and "INC_TYPE_113", are not selected by any dataset. This suggests that the incident types "fire in structure, other than in a

building" (represented by INC_TYPE_112) and "cooking fire" (represented by INC_TYPE_113) appear to have less predictive value for modeling the number of firefighters demanded. Compared to other discrete features, the values of "PROP_USE" and "ACT_TAK1" are selected by more datasets, indicating that their values have more universality and stability in contributing to the model training.

Different from the sparse input matrix corresponding to one-hot encoding, each continuous feature exists only as a column in the input matrix, so its contribution to the target value is relatively higher and has been selected by more datasets. Among all the continuous features, "BLDG_ABOVE", "PROP_VAL", "CONT_VAL", "BLDG_AREA", "ARR_TIME", "AWND" and "TAVG" are selected by all datasets. This result shows that these features have a stable impact on target value under different compositions of the dataset and feature selection methods. Moreover, the selection rate of weather-related features reaches 67.5 %, indicating that weather also has an impact on predicting firefighters demand for urban fires. Therefore, weather should be considered as one of the factors in modeling the number of firefighters at the alarm time.

4.2. Results of initial demand prediction models performance

After feature selection, initial demand prediction models were

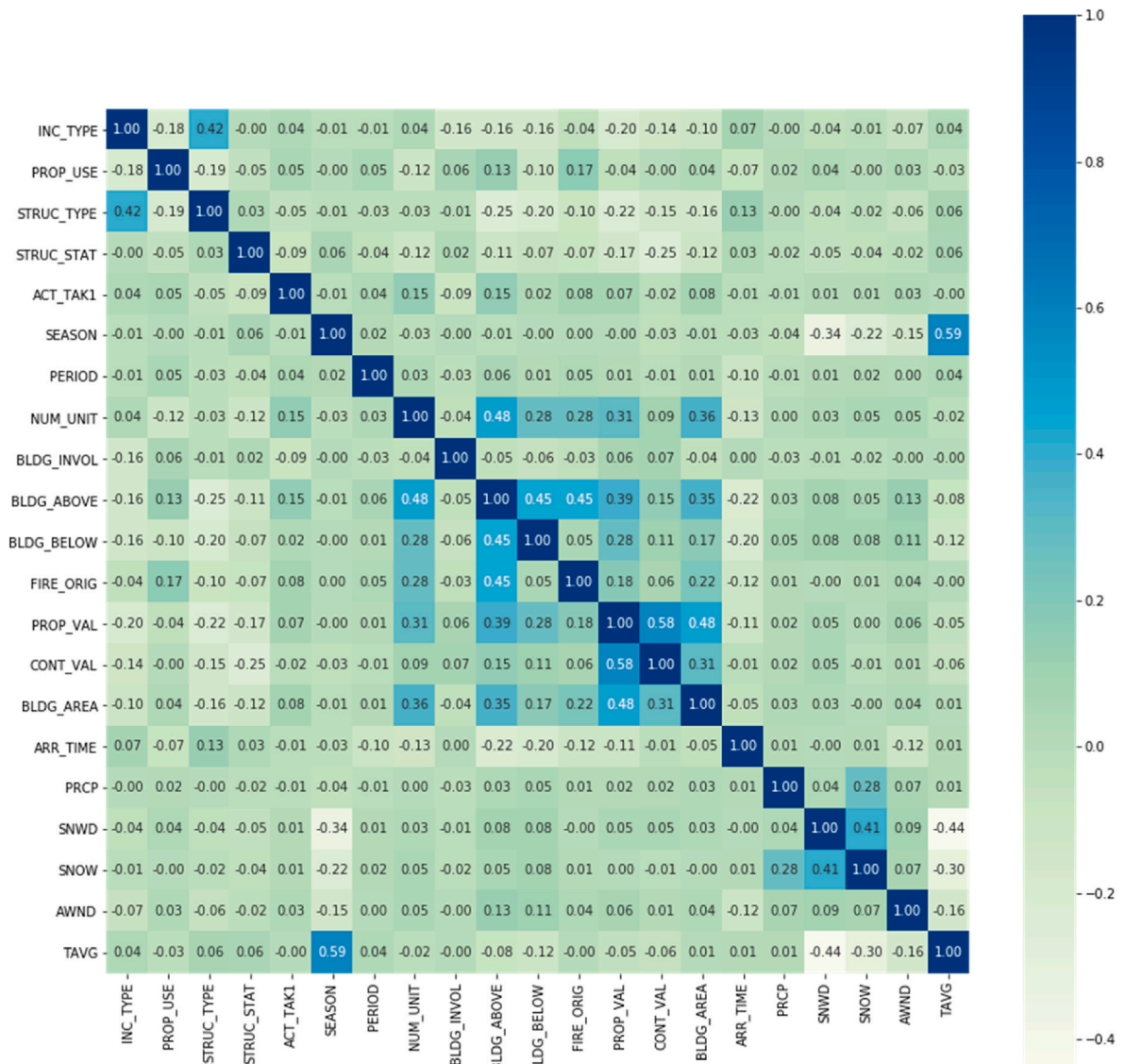


Fig. 5. Test of features' correlation.

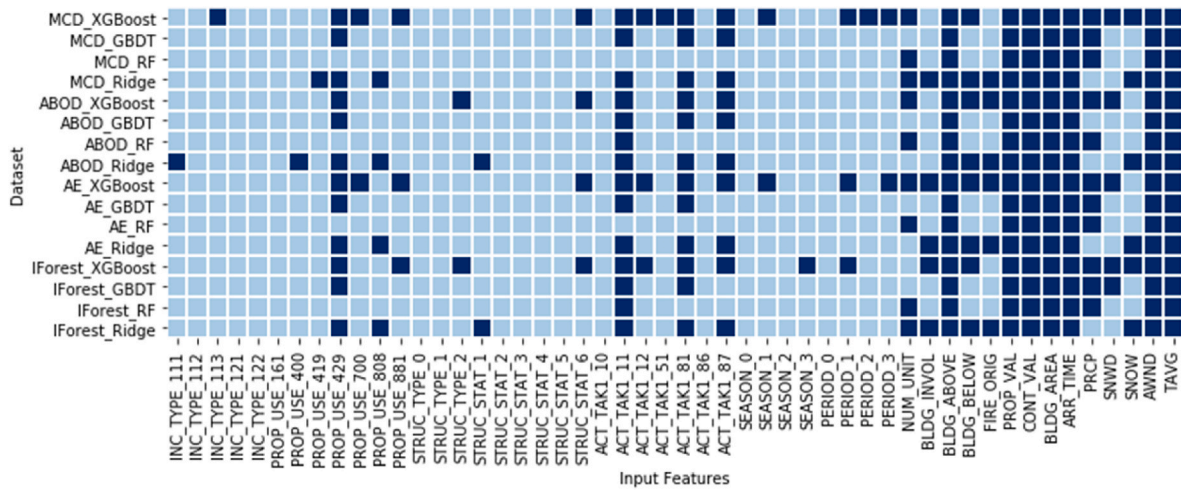


Fig. 6. Results of feature selection on each dataset.

constructed based on the 16 datasets. These models implemented using the sequential model in Keras, which is the most widely used tool for practical applications of ANN. The performance of them is shown in Table 9. MAE values are used to evaluate the effectiveness of the regression model, while PSI, PA and PSR values are evaluation indicators set for the firefighters demand scenarios in urban fires, representing Proportion of Severe Inadequate, Proportion of Acceptable and Proportion of Severe Redundancy, respectively. In this context, lower values for MAE, PSI, and PSR are better as they indicate superior model performance. On the other hand, a higher PA value is better as it denotes a higher proportion of correct predictions made by the model.

Each anomaly detection method was combined with four feature selection methods to obtain four datasets, and the indicators obtained after training the initial demand prediction model on each dataset were compared separately. The optimal results for these indicators under each anomaly detection method are highlighted in bold in Table 9.

For MCD, the initial demand prediction model on MCD_XGBoost dataset has the best performance in terms of MAE, PA and PSR indicators. The MCD_Ridge dataset has the lowest PSI value, but the performance of other indicators is general. In the same way, in the datasets corresponding to the IF algorithm, the IF_XGBoost has the best performance except for the PSI indicator, while the IF_GBDT has the lowest PSI value of 0.1960, which is slightly better than the IF_XGBoost by 0.0047. Among the datasets of ABOD, ABOD_XGBoost achieves optimal MAE of

4.3633 and PA of 0.6406. Meanwhile, ABOD_Ridge and ABOD_GBDT achieve the best PSI and PSR values of 0.1874 and 0.1682, a slight improvement over the ABOD_XGBoost. In addition, for the datasets of AE, AE_Ridge performs the best in the remaining three indicators except MAE, while AE_XGBoost has the lowest MAE value of 4.5788.

The results indicate that no initial demand prediction model excels in all four indicators. Therefore, a scoring rule described in Section 3.2.3 is employed to further evaluate the performance of them. The results are shown in Table 10. The columns represent anomaly detection methods, rows represent feature selection methods, and each value represents the score of initial demand prediction model trained on the dataset generated by that combination method. The ranking scores reveal that in datasets constructed by AE for anomaly detection, the performance of initial demand prediction model trained on the feature subset constructed by Ridge is superior to others. In datasets processed by other anomaly detection methods, using XGBoost for feature selection yields better results. To summarize, the initial demand prediction models trained on the datasets generated by MCD_XGBoost, ABOD_XGBoost, AE_Ridge and IForest_XGBoost combination methods win. This combination methods are further proven through the GA optimization process.

4.3. Results of optimization

During the optimization process, genetic algorithm (GA) was applied to optimize winner models structure. This can further improve their effectiveness and select final optimal one. It is noteworthy that the optimization of ANN structure represents a non-convex optimization problem. Consequently, the ANN structure optimized using GA is a local optimal structure typically, and local optimal structure do not necessarily coincide with the global optimal structure. Moreover, identifying the global optimal structure for this problem proves to be particularly challenging. In the GA-ANN model employed in this paper, ten-fold cross-validation was utilized to enhance the stability of the experimental results. Additionally, this paper conducted 30 repeated experiments to minimize the likelihood of the optimization process get stuck in a local optimum. The parameter settings for individual coding, population evolution and probability of mutations are shown in Table 11.

Table 9
Performance of initial demand prediction models.

Anomaly Detection	Dataset	MAE (min)	PSI (min)	PA (max)	PSR (min)
MCD	MCD_XGBoost	4.6007	0.1985	0.6175	0.1840
	MCD_GBDT	4.6720	0.1982	0.6008	0.2010
	MCD_RF	4.7877	0.2010	0.5882	0.2108
	MCD_Ridge	4.6721	0.1953	0.6005	0.2042
ABOD	ABOD_XGBoost	4.3633	0.1887	0.6406	0.1707
	ABOD_GBDT	4.3800	0.1944	0.6374	0.1682
	ABOD_RF	4.4606	0.1922	0.6213	0.1865
	ABOD_Ridge	4.3794	0.1874	0.6330	0.1796
AE	AE_XGBoost	4.5788	0.1994	0.6119	0.1887
	AE_GBDT	4.7018	0.1972	0.5973	0.2054
	AE_RF	4.7658	0.2023	0.5958	0.2019
	AE_Ridge	4.6582	0.1947	0.6179	0.1874
IF	IF_XGBoost	4.6174	0.2007	0.6144	0.1849
	IF_GBDT	4.6969	0.1960	0.6043	0.1997
	IF_RF	4.6915	0.2051	0.6037	0.1912
	IF_Ridge	4.6668	0.1963	0.6056	0.1982

Table 10
Ranking scores of initial demand prediction models.

Dataset	MCD	ABOD	AE	IF
XGBoost	14	14	12	14
GBDT	12	10	8	8
RF	4	5	5	7
Ridge	10	11	15	11

Table 11

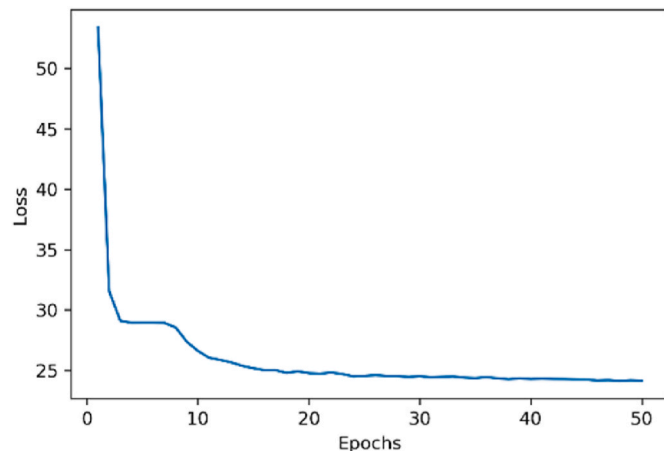
Parameter setting of individual coding, population evolution and probability of mutations in GA.

Parameter		Value		
Individual coding design	max neurons in one layer (MAX_NEURONS)	100		
	min neurons in one layer (MIN_NEURONS)	20		
	max hidden layers (MAX_LAYERS)	3		
	the collection of activation functions (ACTIVATIONS)	[Linear, Sigmoid, hard_Sigmoid, ReLU, Tanh]		
Population evolution design	population size (POP)	10		
	generations (GEN)	5		
	number of individuals selected in Tournament (SIZE)	3		
Probability of mutations	types of mutation modes P_n	mutation within a layer (P_1)	add layers (P_2)	delete layers (P_3)
		0.50	0.25	0.25
	types of mutation within a layer P_{1n}	change the number of neurons (P_{11})	change activation function (P_{12})	change both (P_{13})
		0.50	0.25	0.25
	types of changing the number of neurons P_{11n}	add one (P_{111})	delete one (P_{112})	reinitialize (P_{113})
		0.30	0.30	0.40

Besides, the batch size and epochs were separately set to 10 and 50.

The decrease of loss value in the training process of the optimized ANN is shown in Fig. 7. With the increase of epochs, the loss value gradually converges, which shows that the optimized ANN has gradually approaching the optimal structure.

Optimized structures of the winner initial demand prediction models are shown in Table 12. It clearly shows that the demand prediction models on different datasets get the model structures with different depths, and all the optimized models have better performance in MAE. Among them, the most notable is the demand prediction model based on the ABOD_XGBoost dataset. It has the lowest MAE of 4.1007, which is 0.2622 lower than before.

**Fig. 7.** Optimization iteration process of ANN structure based on GA.**Table 12**

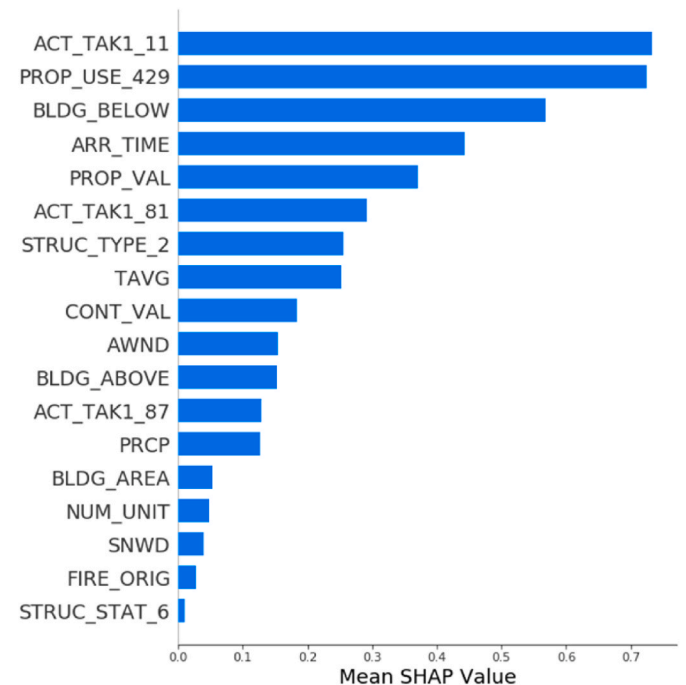
Optimized structure of GA-ANN demand prediction model.

Dataset	Depth	Structure of Each Layer	MAE	Pre_MAE
MCD_GBDT	1	Dense (46, sigmoid)	4.5829	4.6007
	2	Dense (81, sigmoid)		
	3	Dense (77, relu)		
	4	Dense (30, linear)		
	5	Dense (1)		
ABOD_XGBoost	1	Dense (60, sigmoid)	4.1007	4.3629
	2	Dense (87, linear)		
	3	Dense (34, sigmoid)		
	4	Dense (1)		
AE_Ridge	1	Dense (27, tanh)	4.5804	4.6582
	2	Dense (36, tanh)		
	3	Dense (1)		
IF_XGBoost	1	Dense (68, hard_sigmoid)	4.5495	4.6174
	2	Dense (38, sigmoid)		
	3	Dense (63, linear)		
	4	Dense (42, sigmoid)		
	5	Dense (1)		

4.4. Analysis of model interpretability

In addition, this paper provides interpretability analysis of the demand prediction model. Interpretability analysis enables decision makers to further understand how the model makes predictions and enhance its transparency, thus improve the usability of the model [71]. In other words, an objective evaluation of how each feature affects firefighter demand can provide critical insights into the fire commander's dispatch decision process revealed by a data-driven approach.

In this paper, Shapley method is used for interpretability analysis. In terms of overall interpretability, Shapley values of each feature in the model are calculated and sorted according to their average values to obtain important features. Features with larger Shapley values play a more important role in prediction [72]. As can be seen from Fig. 8, ACT_TAK1_11 (Whether the Primary Action Taken is Extinguishment by fire service personnel), PROP_USE_429 (Whether the building is used as a Multifamily dwelling), BLDG_BELOW (Building Height: Stories Below Grade), ranked in the top three. A detailed interpretability analysis of

**Fig. 8.** Importance analysis of factors affecting firefighters demand.

these three features is presented below.

In Fig. 9, blue represents a low feature value, red represents a high feature value, and a positive SHAP value represents a positive contribution and a negative value represents a negative contribution. As can be seen in the figure, ACT_TAK1_11 shows a high positive contribution, indicating that when receiving a fire alarm, if it is decided that the primary task is to dispatch firefighters to extinguish the fire, the positive contribution will be made to the demand of firefighters (because some urban fires that are not serious, the fire department will only provide guidance to the alarm personnel on the use of fire extinguishers or the use of fire hydrants). PROP_USE_429 shows a high value positive contribution to the firefighters demand when building is a Multifamily dwelling (multifamily dwelling are more densely populated, resulting in greater casualties and property damage from the fire). And BLDG_BELOW shows a high positive contribution, indicating that the lower the floor, the more positive contribution it produces to firefighters demand (the lower the floor, the easier it is for firefighters to enter the scene to fight the fire, and for building with high floors, where firefighters are not easy to enter the scene, they tend to use more professional equipment). Obviously, these interpretability analyses are consistent with the realistic scenario. Understanding the importance of these features and their contribution to the demand of firefighters can help fire commanders to take corresponding countermeasures in different scenarios in time, and further improve the usability of demand prediction model.

4.5. Comparison results

4.5.1. Effect of genetic algorithm optimization

As mentioned in Section 4.3, MAE values of all models significantly improved after optimization using genetic algorithm. Table 13 presents a comparison of all indicators before and after GA structure optimization for the four winner models in the initial demand prediction model construction.

For the IF_XGBoost dataset, all evaluation indicators improve after GA optimization process. For the other datasets, the results of indicators show that while PSI and PSR values fluctuate slightly, PA and MAE values improve significantly after optimization. From the results in

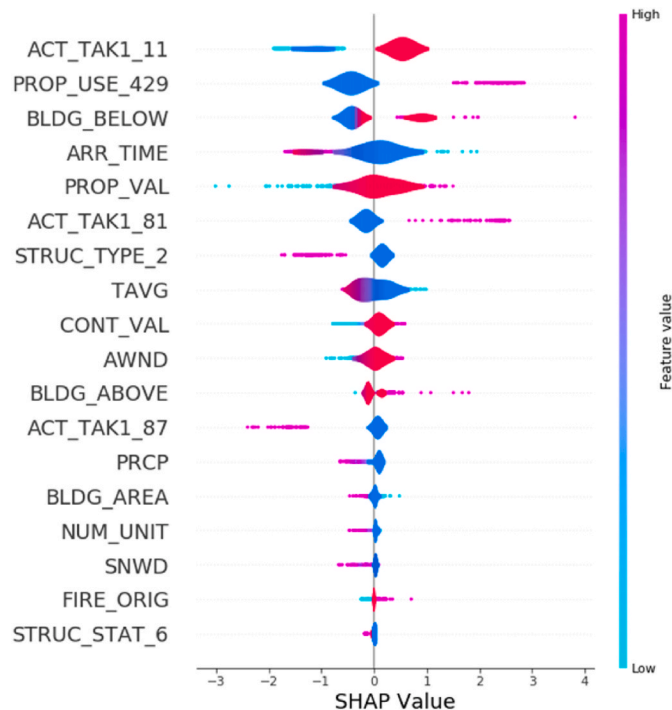


Fig. 9. SHAP values distribution of factors affecting firefighters demand.

Table 13

Effect of genetic algorithm to optimize ANN structure.

Dataset	Demand Prediction Model	MAE (min)	PSI (min)	PA (max)	PSR (min)
MCD_XGBoost	ANN	4.6007	0.1985	0.6175	0.1840
	GA-ANN	4.5829	0.2057	0.6267	0.1676
ABOD_XGBoost	ANN	4.3629	0.1824	0.6387	0.1789
	GA-ANN	4.1007	0.1833	0.6747	0.1420
AE_Ridge	ANN	4.6582	0.1947	0.6179	0.1874
	GA-ANN	4.5804	0.1849	0.6261	0.1890
IF_XGBoost	ANN	4.6174	0.2007	0.6144	0.1849
	GA-ANN	4.5495	0.1982	0.6257	0.1761

Section 4.3, it can be seen that the optimized GA-ANN demand prediction model is obtained on ABOD_XGBoost dataset, which is now illustrated as an example. Although PSI value increases slightly by 0.0009, MAE value decreases by 0.2622, PA value increases by almost 6 %, and PSR value decreases more than 20 %. This indicates that the optimization of artificial neural network structure by genetic algorithm is effective.

4.5.2. Comparison results with popular regression models

In order to verify the effectiveness of GA-ANN demand prediction model proposed in this paper, this section compares it with other mainstream machine learning models. Among these models used for comparison, the basic models are Decision Tree (DT), Linear Regression (LR), Support Vector Machines (SVM), Lasso Regression, Ridge Regression, ElasticNet Regression and K-Nearest Neighbor (KNN), while the ensemble models include eXtreme Gradient Boosting (XGBoost), Adaptive Boosting (AdaBoost), Gradient Boosting Decision Tree (GBDT), Light Gradient Boosting Machine (LightGBM), Extra Tree (ET), Random Forest (RF) and Bagging. Because the optimized GA-ANN demand prediction model is obtained on ABOD_XGBoost dataset, comparison models were also constructed on it, and relevant parameters were adjusted. The indicators of each model after ten-fold cross validation are shown in Table 14.

MAE value of the GA-ANN model proposed in this paper is only 4.1007, which is 0.3011 lower than the second-best (GBDT) and 1.9234 lower than the worst (DT), showing the best result among all models compared. Similarly, the proposed model achieves a PA value of 0.6747, indicating that nearly 70 % of the predicted results are acceptable and in line with expectations, which is much better than other models.

On the other hand, although AdaBoost has the lowest PSI value of 0.1505, its PSR value of 0.2789 is higher than any other model. Similarly, SVM has a surprisingly low PSR of 0.1139, but its PSI is the highest

Table 14

Comparison results with popular regression models.

Models		MAE (min)	PSI (min)	PA (max)	PSR (min)
Proposed Model	GA- ANN	4.1007	0.1833	0.6747	0.1420
Base Models	DT	6.0241	0.2234	0.5468	0.2297
	LR	4.4291	0.1944	0.6292	0.1764
	SVM	4.3380	0.2307	0.6554	0.1139
	Lasso	4.4368	0.1928	0.6302	0.1770
	Ridge	4.4292	0.1941	0.6295	0.1764
	ElasticNet	4.7584	0.2017	0.5841	0.2142
	KNN	4.7487	0.1976	0.6106	0.1918
Ensemble Models	XGBoost	4.7266	0.1988	0.6071	0.1940
	AdaBoost	4.6995	0.1505	0.5705	0.2789
	GBDT	4.4018	0.1922	0.6308	0.1770
	LightGBM	4.5344	0.1950	0.6302	0.1748
	ET	4.5687	0.1878	0.6138	0.1985
	RF	4.5100	0.1802	0.6210	0.1988
	Bagging	4.6253	0.1903	0.6122	0.1975

of all models at 0.2307. These results highlight the instability of these models in predicting firefighters demand. The proposed model is not the best in terms of PSI and PSR indicators, but it also ranks in the top three of all compared models. Specifically, the GA-ANN model has a PSI value of 0.1833, which is just behind 0.1505 for AdaBoost and 0.1802 for RF. Whereas, its PSR has a value of 0.1420, which is second only to SVM model of 0.1139, and improves by 18.59 % in comparison with the third best LightGBM. In conclusion, the proposed GA-ANN model has the best overall prediction performance, which is a robust and reliable tool to predict the demand for firefighters.

4.5.3. Effect of merging weather data

Weather data has not been applied in previous research on predicting the demand for firefighters. To verify the validity of weather data, firstly, this part deleted all weather-related features in the four datasets obtained from anomaly detection. Then the feature subset construction and model training were repeated. When constructing the feature subset, optimal feature selection method for each anomaly detection method shown in Table 10 was used. The optimized structure of each winner initial demand prediction model shown in Table 12 was used for model training. The above operation ensures that the other parts of the experiment remain consistent except for weather data. Finally, the performance of each model is shown in Table 15.

As can be seen from Table 15, all four indicators in the datasets obtained by AE_Ridge and IF_XGBoost combination methods were significantly improved with the addition of weather data. Although the PSI values of MCD_XGBoost and ABOD_XGBoost have a small increase of 0.0028 and 0.0056 respectively after the addition of weather data, all other indicators have been optimized. Especially the ABOD_XGBoost dataset, which is the best performing one in Section 4.3, its MAE value decreases by 0.2072 and PSR value decreases by 12.45 %.

To sum up, models with weather data exhibit lower MAE, lower PSR and higher PA across all datasets. While there is a slight increase of PSI in MCD_XGBoost and ABOD_XGBoost datasets, this is accompanied by a decrease in PSR and an increase in PA, which shows better performance overall. Therefore, weather has a very positive impact on predicting the number of firefighters demanded for urban fires. The merging of weather data can effectively enhance the accuracy of predictions, making it a crucial factor to consider.

5. Conclusion

This paper proposes a novel historical data-based method for predicting firefighters demand in urban fires, combining fire data from NFIRS and weather data from GHCN-D to create a fused dataset. Based on this, four anomaly detection methods are utilized for data pre-processing, and further the method for constructing the firefighters demand prediction model is proposed. The construction process includes feature selection which considered the effect of anomaly detection, initial demand prediction model construction and GA-based demand prediction model optimization. The results show that the method proposed in this paper is more accurate than other mainstream machine learning methods. It will better support the urban fire departments in making decisions on firefighters demand prediction issues.

There are four innovations in this paper. Firstly, it considers the use of real historical data to predict the number of firefighters demanded for urban fires, which provide a data-driven objective view of firefighter demand prediction for urban fires. Secondly, this paper proposes a dataset construction method that combines anomaly detection and feature selection. There are 16 datasets which are built by combination of four anomaly detection methods with four feature selection methods respectively. The characteristics of different methods are utilized to capture anomalous patterns in the data, in order to reduce the impact of data noise and select effective features. This helps to improve the performance of subsequent demand prediction models. Thirdly, this paper proposes three demand prediction model evaluation indicators,

Table 15

Effect of merging weather data.

Dataset	Model	MAE	PSI	PA	PSR
		(min)	(min)	(max)	(min)
MCD_XGBoost without weather	GA-ANN	4.5959	0.2029	0.6191	0.1780
MCD_XGBoost		4.5829	0.2057	0.6267	0.1676
ABOD_XGBoost without weather	GA-ANN	4.3079	0.1777	0.6602	0.1622
ABOD_XGBoost		4.1007	0.1833	0.6747	0.1420
AE_Ridge without weather	AE_Ridge	4.6637	0.1900	0.6103	0.1997
AE_Ridge		4.5804	0.1849	0.6261	0.1890
IF_XGBoost without weather	IF_XGBoost	4.6354	0.2048	0.6160	0.1792
IF_XGBoost		4.5495	0.1982	0.6257	0.1761

Proportion of Severe Inadequacy (PSI), Proportion of Acceptability (PA) and Proportion of Severe Redundancy (PSR). These indicators combine the characteristics of the firefighters demand problem in the urban fire field to evaluate the performance of the demand prediction model, and the evaluation results provide a more comprehensive insight into the accuracy of the model. Finally, this paper proposes a GA-based structure optimization method for demand prediction model. During the optimization of the model structure, the optimized number of hidden layers, number of neurons per layer and activation function per layer of ANN are determined, and parameters are optimized based on optimized structure. This will greatly ensure the accuracy of the demand prediction model, and finally provide an accurate prediction of the number of firefighters demanded for urban fires.

This study has made strides toward data-based firefighters demand in urban fires. There are a few more things that could be done to make it perform better, such as.

- (1) Collecting more data related to fire accidents to replace or combine with current data, thereby improving the quality of the raw data.
- (2) Using data from each state or even much smaller areas to build different demand prediction models that better adapt to local characteristics.
- (3) Continue to improve the accuracy of the model prediction results by combining deep learning methods

In conclusion, while this study represents a step forward in data-based firefighters demand, there are many exciting opportunities for further research and development in this area.

CRedit authorship contribution statement

Chen-yue Zhang: Writing – original draft, Validation, Software, Investigation, Data curation. **Rui Zhao:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization. **Ning Wang:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Xin Nie:** Writing – original draft, Software, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the National Key R&D Program of China (2021YFC3300200), the National Natural Science Foundation of China (Nos. 72304059) and Dalian Science & Technology Innovation Fund Project (2022JJ12GX 012).

References

- [1] A. Asgary, A. Sadeghi Naini, J. Levy, Modeling the risk of structural fire incidents using a self-organizing map, *Fire Saf. J.* 49 (2012) 1–9, <https://doi.org/10.1016/j.firesaf.2011.12.007>.
- [2] T. Purvis, C. Davis, M. Marks, G. Scott, S. McGehee, I. Gardett, S. Sangaraju, C. Olola, Identifying Working Structure Fires Using a Standardized Fire Dispatch Protocol System, vol. 8, 2020, p. 4.
- [3] A. Asgary, A. Ghaffari, J. Levy, Spatial and temporal analyses of structural fire incidents and their causes: a case of Toronto, Canada, *Fire Saf. J.* 45 (2010) 44–57, <https://doi.org/10.1016/j.firesaf.2009.10.002>.
- [4] J. Parente, M.G. Pereira, Structural fire risk: the case of Portugal, *Sci. Total Environ.* 573 (2016) 883–893, <https://doi.org/10.1016/j.scitotenv.2016.08.164>.
- [5] National Fire Protection Association (NFPA), NFPA 1720, 2020 [Online]. Available, <http://www.nfpa.org>.
- [6] T. Parhizkar, S. Hogenboom, J.E. Vinnem, I.B. Utne, Data driven approach to risk management and decision support for dynamic positioning systems, *Reliab. Eng. Syst. Saf.* 201 (2020) 106964, <https://doi.org/10.1016/j.res.2020.106964>.
- [7] C.J. Pérez-González, M. Colebrook, J.L. Roda-García, C.B. Rosa-Remedios, Developing a data analytics platform to support decision making in emergency and security management, *Expert Syst. Appl.* 120 (2019) 167–184, <https://doi.org/10.1016/j.eswa.2018.11.023>.
- [8] M.C. Arienti, S.G. Cumming, S. Boutin, Empirical models of forest fire initial attack success probabilities: the effects of fuels, anthropogenic linear features, fire weather, and management, *Can. J. For. Res.* 36 (2006) 3155–3166, <https://doi.org/10.1139/x06-188>.
- [9] T. Kox, C. Lüder, Impacts as Triggers for weather-related decision making: observations at the Berlin fire Brigade Control and dispatch center, *Int J. Disaster Risk Sci* 12 (2021) 610–615, <https://doi.org/10.1007/s13753-021-00356-4>.
- [10] A. Paudel, D.L. Martell, D.G. Woolford, Factors that affect the timing of the dispatch of initial attack resources to forest fires in northeastern Ontario, Canada, *Int. J. Wildland Fire* 28 (2019) 15, <https://doi.org/10.1071/WF18058>.
- [11] J.-L. Wybo, FMIS: a decision support system for forest fire prevention and fighting, *IEEE Trans. Eng. Manag.* 45 (1998) 127–131, <https://doi.org/10.1109/17.669745>.
- [12] O. Njå, M. Svela, A review of competencies in tunnel fire response seen from the first responders' perspectives, *Fire Saf. J.* 97 (2018) 137–145, <https://doi.org/10.1016/j.firesaf.2017.05.005>.
- [13] R. Gazzard, J. McMorrow, J. Aylen, Wildfire policy and management in England: an evolving response from Fire and Rescue Services, forestry and cross-sector groups, *Phil. Trans. R. Soc. B.* 371 (2016) 20150341, <https://doi.org/10.1098/rstb.2015.0341>.
- [14] S. Rezaeifam, E. Ergen, H.M. Günaydin, Fire emergency response systems information requirements' data model for situational awareness of responders: a goal-directed task analysis, *J. Build. Eng.* 63 (2023) 105531, <https://doi.org/10.1016/j.jobbe.2022.105531>.
- [15] E. Yeboah, I. Sarfo, E.K. Nunoo, V. Kudoh, N. Shang, F.K. Addai, S.O.Y. Amankwah, E.A.G. Kedjanyi, GIS-based emergency fire response for minimization of fire outbreaks in the greater accra metropolis, Ghana, *JGEESI* (2021) 30–45, <https://doi.org/10.9734/jgeesi/2021/v25i530286>.
- [16] C. Andréll, J. Dankiewicz, L. Todorova, K. Olanders, S. Ullén, H. Friberg, Firefighters as first-responders in out-of-hospital cardiac arrest – a retrospective study of a time-gain selective dispatch system in the Skåne Region, Sweden, *Resuscitation* 179 (2022) 131–140, <https://doi.org/10.1016/j.resuscitation.2022.08.012>.
- [17] H. Yoshioka, K. Himoto, K. Kagiya, Large urban fires in Japan: history and management, *Fire Technol.* 56 (2020) 1885–1901, <https://doi.org/10.1007/s10694-020-00960-8>.
- [18] D.Y. Kim, K.R. Persad, N.A. Khwaja, S. Chi, Assessment of staffing needs for construction inspection, *KSCCE J. Civ. Eng.* 20 (2016) 2598–2603, <https://doi.org/10.1007/s12205-016-0817-6>.
- [19] Y. Li, T.R.B. Taylor, G.B. Dadi, R.E. Sturgill, Modeling long-term highway staffing requirements for state transportation agencies, *Transport. Res. Rec.* 2612 (2017) 76–84, <https://doi.org/10.3141/2612-09>.
- [20] C.Y. Pang, J.M. Swint, Forecasting staffing needs for productivity management in hospital laboratories, *J. Med. Syst.* 9 (1985) 365–377, <https://doi.org/10.1007/BF00992574>.
- [21] W. Chen, H. Guo, K.-L. Tsui, A new medical staff allocation via simulation optimisation for an emergency department in Hong Kong, *Int. J. Prod. Res.* 58 (2020) 6004–6023, <https://doi.org/10.1080/00207543.2019.1665201>.
- [22] L.V. Green, P.J. Kolesar, Anniversary paper: improving emergency responsiveness with management science, *Manag. Sci.* 50 (2004) 1001–1014, <https://doi.org/10.1287/mnsc.1040.0253>.
- [23] P.J. Kolesar, OM forum – some lessons on operations management model implementation drawn from the RAND fire project, *MS* 14 (2012) 1–6, <https://doi.org/10.1287/msom.1110.0367>.
- [24] G. Carter, E. Ignall, W. Walker, A simulation model of the New York city fire department: its use in deployment analysis, in: *Proceedings of the 6th Conference on Winter Simulation*, Association for Computing Machinery, New York, NY, USA, 1973, pp. 353–370, <https://doi.org/10.1145/800293.811589>.
- [25] G. Carter, E. Ignall, A simulation model of fire department operations: design and preliminary results, *IEEE Trans. Syst. Sci. Cybern.* 6 (1970) 282–293, <https://doi.org/10.1109/TSSC.1970.300303>.
- [26] E. Ignall, G. Carter, K. Rider, An algorithm for the initial dispatch of fire companies, *Manag. Sci.* 28 (1982) 366–378, <https://doi.org/10.1287/mnsc.28.4.366>.
- [27] A.J. Swersey, A markovian decision model for deciding how many fire companies to dispatch, *Manag. Sci.* 28 (1982) 352–365, <https://doi.org/10.1287/mnsc.28.4.352>.
- [28] A. Mukhopadhyay, Z. Wang, Y. Vorobeychik, A Decision Theoretic Framework for Emergency Responder Dispatch, *Industrial Applications*, 2018, p. 588.
- [29] A. Mukhopadhyay, G. Pettet, C. Samal, A. Dubey, Y. Vorobeychik, An online decision-theoretic pipeline for responder dispatch, in: *Proceedings of the 10th ACM/IEEE International Conference on Cyber-Physical Systems*, 2019, pp. 185–196, <https://doi.org/10.1145/3302509.3311055>.
- [30] D. Usanov, P.M. van de Ven, R.D. van der Mei, Dispatching fire trucks under stochastic driving times, *Comput. Oper. Res.* 114 (2020) 104829, <https://doi.org/10.1016/j.cor.2019.104829>.
- [31] M. Armsbarger, J. Goldstein, C. Kelling, G. Korkmaz, S. Keller, Modeling response time to structure fires, *Am. Statistician* 75 (2021) 92–100, <https://doi.org/10.1080/00031305.2019.1695664>.
- [32] C.J. Dunn, M.P. Thompson, D.E. Calkin, A framework for developing safe and effective large-fire response in a new fire management paradigm, *For. Ecol. Manag.* 404 (2017) 184–196, <https://doi.org/10.1016/j.foreco.2017.08.039>.
- [33] T. Ishida, S. Hoshino, Implementation of a volunteer fire corps activity support system during fires, *IJWIS* 15 (2019) 385–401, <https://doi.org/10.1108/IJWIS-09-2018-0068>.
- [34] S. Basu, S. Roy, S. DasBit, A post-disaster demand forecasting system using principal component regression analysis and case-based reasoning over smartphone-based DTN, *IEEE Trans. Eng. Manag.* 66 (2019) 224–239, <https://doi.org/10.1109/TEM.2018.2794146>.
- [35] P. Avesani, A. Perini, F. Ricci, Interactive case-based planning for forest fire management, *Appl. Intell.* 13 (2000) 41–57, <https://doi.org/10.1023/a:1008327312956>.
- [36] L. Fei, Y. Wang, Demand prediction of emergency materials using case-based reasoning extended by the Dempster-Shafer theory, *Soc. Econ. Plann. Sci.* 84 (2022) 101386, <https://doi.org/10.1016/j.seps.2022.101386>.
- [37] A. Safarishahrbi, Workforce forecasting models: a systematic review, *J. Forecast.* 37 (2018) 739–753, <https://doi.org/10.1002/for.2541>.
- [38] H. Elkhooly, R. Ead, A. Hammad, S. AbouRizk, Data mining for forecasting labor resource requirements: a case study of project management staffing requirements, *International Journal of Construction Management* (2022) 1–12, <https://doi.org/10.1080/15623599.2022.2112898>.
- [39] S. Yuan, Q. Qi, E. Dai, Y. Liang, Human resource planning and configuration based on machine learning, *Comput. Intell. Neurosci.* 2022 (2022) 1–6, <https://doi.org/10.1155/2022/3605722>.
- [40] W. Kumwilaisak, S. Phikulngoen, J. Piriayataravet, N. Thatphithakkul, C. Hansakunbuntheung, Adaptive call center workforce management with deep neural network and reinforcement learning, *IEEE Access* 10 (2022) 35712–35724, <https://doi.org/10.1109/ACCESS.2022.3160452>.
- [41] S.L. Rose-Pehrson, S.J. Hart, T.T. Street, F.W. Williams, M.H. Hammond, Early warning fire detection system using a probabilistic neural network, *Fire Technol.* 39 (2003) 147–171, <https://doi.org/10.1023/A:1024260130050>.
- [42] D. Xia, Fire risk evaluation model of high-rise buildings based on multilevel BP neural network, in: *Fourth International Conference on Fuzzy Systems and Knowledge Discovery (FSKD 2007)*, IEEE, Haikou, China, 2007, pp. 436–441, <https://doi.org/10.1109/FSKD.2007.291>.
- [43] E. Gelenbe, S. Timotheou, D. Nicholson, S. Building, *Random Neural Network for Emergency Management*, 2010.
- [44] N. Cheng, X. Song, Evaluation research on emergency management capability of college accidents based on improved LM-RBF neural network, in: *Proceedings of the Third International Conference on Economic and Business Management (FEBM 2018)*, Atlantis Press, Hohhot, China, 2018, <https://doi.org/10.2991/feb-18.2018.78>.
- [45] D. Jia, Z. Wu, Intelligent evaluation system of government emergency management based on BP neural network, *IEEE Access* 8 (2020) 199646–199653, <https://doi.org/10.1109/ACCESS.2020.3032462>.
- [46] A. Sadeghi-Naini, A. Asgary, Modeling number of firefighters responding to an incident using artificial neural networks, *International Journal of Emergency Services* 2 (2013) 104–118, <https://doi.org/10.1108/IJES-03-2012-0001>.
- [47] D.P. Kingma, J. Ba, Adam: a method for stochastic optimization. <https://doi.org/10.48550/arXiv.1412.6980>, 2017.
- [48] M.D. Zeiler, ADADELTA: an adaptive learning rate method. <https://doi.org/10.48550/arXiv.1212.5701>, 2012.
- [49] M.C. Mukkamala, M. Hein, Variants of RMSProp and Adagrad with logarithmic regret bounds. <https://doi.org/10.48550/arXiv.1706.05507>, 2017.
- [50] Y. Zhang, Z. Jin, Y. Chen, Hybridizing grey wolf optimization with neural network algorithm for global numerical optimization problems, *Neural Comput. Appl.* 32 (2020) 10451–10470, <https://doi.org/10.1007/s00521-019-04580-4>.
- [51] X. Lu, A human resource demand forecasting method based on improved BP algorithm, *Comput. Intell. Neurosci.* 2022 (2022) 1–9, <https://doi.org/10.1155/2022/3534840>.

- [52] Q. Zhang, C. Wang, Using genetic algorithm to optimize artificial neural network: a case study on earthquake prediction, in: 2008 Second International Conference on Genetic and Evolutionary Computing, IEEE, Jinzhou, China, 2008, pp. 128–131, <https://doi.org/10.1109/WGEC.2008.96>.
- [53] L. Wang, A hybrid genetic algorithm–neural network strategy for simulation optimization, *Appl. Math. Comput.* 170 (2005) 1329–1343, <https://doi.org/10.1016/j.amc.2005.01.024>.
- [54] Y. Wang, C. Chen, X. Yan, Structure and weight optimization of neural network based on CPA-MLR and its application in naphtha dry point soft sensor, *Neural Comput. Appl.* 22 (2013) 75–82, <https://doi.org/10.1007/s00521-012-1044-9>.
- [55] L. Li, W. Xu, H. Yu, Character-level neural network model based on Nadam optimization and its application in clinical concept extraction, *Neurocomputing* 414 (2020) 182–190, <https://doi.org/10.1016/j.neucom.2020.07.027>.
- [56] H.-P. Kriegel, M. S. Hubert, A. Zimek, Angle-based outlier detection in high-dimensional data, in: *Proceeding of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining - KDD 08*, ACM Press, Las Vegas, Nevada, USA, 2008, p. 444, <https://doi.org/10.1145/1401890.1401946>.
- [57] H. Xu, C. Caramanis, S. Sanghavi, Robust PCA via outlier pursuit, *IEEE Trans. Inf. Theor.* 58 (2012) 3047–3064, <https://doi.org/10.1109/TIT.2011.2173156>.
- [58] N. Chitradevi, K. Baskaran, V. Palanisamy, D. Aswini, Designing an efficient PCA based data model for wireless sensor networks, in: *Proceedings of the 1st International Conference on Wireless Technologies for Humanitarian Relief - ACWR '11*, ACM Press, Amritapuri, Kollam, Kerala, India, 2011, p. 147, <https://doi.org/10.1145/2185216.2185262>.
- [59] M. Sakurada, T. Yairi, Anomaly detection using autoencoders with nonlinear dimensionality reduction, in: *Proceedings of the MLSDA 2014 2nd Workshop on Machine Learning for Sensory Data Analysis - MLSDA'14*, ACM Press, Gold Coast, Australia QLD, Australia, 2014, pp. 4–11, <https://doi.org/10.1145/2689746.2689747>.
- [60] Y. Liu, Z. Li, C. Zhou, Y. Jiang, J. Sun, M. Wang, X. He, Generative adversarial active learning for unsupervised outlier detection, *IEEE Trans. Knowl. Data Eng.* (2019) 1, <https://doi.org/10.1109/TKDE.2019.2905606>, 1.
- [61] A. Lazarevic, V. Kumar, Feature bagging for outlier detection, in: *Proceeding of the Eleventh ACM SIGKDD International Conference on Knowledge Discovery in Data Mining - KDD '05*, ACM Press, Chicago, Illinois, USA, 2005, p. 157, <https://doi.org/10.1145/1081870.1081891>.
- [62] F.T. Liu, K.M. Ting, Z.-H. Zhou, Isolation forest, in: 2008 Eighth IEEE International Conference on Data Mining, IEEE, Pisa, Italy, 2008, pp. 413–422, <https://doi.org/10.1109/ICDM.2008.17>.
- [63] V.F. Rodriguez-Galiano, J.A. Luque-Espinar, M. Chica-Olmo, M.P. Mendes, Feature selection approaches for predictive modelling of groundwater nitrate pollution: an evaluation of filters, embedded and wrapper methods, *Sci. Total Environ.* 624 (2018) 661–672, <https://doi.org/10.1016/j.scitotenv.2017.12.152>.
- [64] I. Guyon, A. Elisseeff, An introduction to variable and feature selection, *J. Mach. Learn. Res.* 3 (2003) 1157–1182.
- [65] C.G. Chua, A.T.C. Goh, A hybrid Bayesian back-propagation neural network approach to multivariate modelling, *Int. J. Numer. Anal. Methods GeoMech.* 27 (2003) 651–667, <https://doi.org/10.1002/nag.291>.
- [66] J. Chen, Y. Chen, H. Zheng, S. Shen, S. Yu, D. Zhang, Q. Xuan, MGA: Momentum Gradient Attack on Network, 2020. <http://arxiv.org/abs/2002.11320>.
- [67] A. Camero, J. Toutouh, E. Alba, Random error sampling-based recurrent neural network architecture optimization, *Eng. Appl. Artif. Intell.* 96 (2020) 103946, <https://doi.org/10.1016/j.engappai.2020.103946>.
- [68] M.A.J. Idrissi, H. Ramchoun, Y. Ghanou, M. Ettaouil, Genetic algorithm for neural network architecture optimization, in: 2016 3rd International Conference on Logistics Operations Management (GOL), IEEE, Fez, Morocco, 2016, pp. 1–4, <https://doi.org/10.1109/GOL.2016.7731699>.
- [69] S. Mirjalili, Genetic algorithm, in: *Evolutionary Algorithms and Neural Networks*, Springer International Publishing, Cham, 2019, pp. 43–55, https://doi.org/10.1007/978-3-319-93025-1_4.
- [70] Z. Arabasadi, R. Alizadehsani, M. Roshanzamir, H. Moosaei, A.A. Yarifard, Computer aided decision making for heart disease detection using hybrid neural network-Genetic algorithm, *Comput. Methods Progr. Biomed.* 141 (2017) 19–26, <https://doi.org/10.1016/j.cmpb.2017.01.004>.
- [71] S. Nazar, J. Yang, X.-E. Wang, K. Khan, M.N. Amin, M.F. Javed, F. Althoei, M. Ali, Estimation of strength, rheological parameters, and impact of raw constituents of alkali-activated mortar using machine learning and SHapely Additive exPlanations (SHAP), *Construct. Build. Mater.* 377 (2023) 131014, <https://doi.org/10.1016/j.conbuildmat.2023.131014>.
- [72] C.B. Arachchilage, G. Huang, C. Fan, W.V. Liu, Forecasting unconfined compressive strength of calcium sulfoaluminate cement mixtures using ensemble machine learning techniques integrated with shapely-additive explanations, *Construct. Build. Mater.* 409 (2023) 134083, <https://doi.org/10.1016/j.conbuildmat.2023.134083>.